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The University of Chicago Booth School of Business

Examining Capital Bikeshare Rental Counts from Various Predictors

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Honor Code

We pledge our honor that we have not violated the Honor Code during the preparation of this assignment.

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1 Motivation and Executive Summary

Capital Bikeshare is a popular bikesharing system that operates primarily operates in Washington D.C. In 2024 alone, Capital Bikeshare recorded more than six million rides across more than 700 stations throughout the city.

As operations for Capital Bikeshare continue to grow, it is important that they allocate their time and finances to effectively optimize the number of bike rentals in Washington D.C. As a result, Capital Bikeshare may be interested in understanding any inherent tendencies in their consumers; for example, the weather and temperature of a particular day may play an important role in influencing the number of bike rentals.

To incentivize frequent bike rentals, Capital Bikeshare also offers several options that the customer can select from when renting a bike, which are provided below:

- Single Ride - The customer pays a flat rate of $1 for unlocking the bike, and then chooses from either a classic bike or an electronic bike (ebike). The rates are the following: $0.05/min for a classic bike, $0.15/min for an ebike.
- Day Pass - The customer pays a flat rate of $8/day. For the classic bike, the first 45 minutes are free, and afterwards the customer pays $0.05/min. For the ebike the customer pays $0.10/min.
- Capital Bikeshare Membership - The customer pays an annual rate of $95, and the rates for both the classic bike and the ebike are the same as the day pass.

As such, Capital Bikeshare may be interested in evaluating the popularity of their membership program. Specifically, they may be interested in the number of bikes rented by single riders/day pass customers versus the number of bikes rented by Capital Bikeshare members.

Thus, the primary objective of this analysis is to investigate how various factors, such as weather, affect

1. the number of total bikes rented
2. the number of bikes rented by non-members (single riders/day pass customers)
3. the number of bikes rented by Capital Rideshare members

The key findings of our analysis were the following:

- *Naive Linear Regression*:
  - A linear model that predicts the total number of bikes resulted in an in-sample R^2 of 0.800.
  - A linear model that predicts the number of bikes rented by Capital Rideshare members reported an in-sample R^2 of 0.816.
  - A linear model that predicts the number of bikes rented by non-members reported an an in-sample R^2 of 0.688.
- *Cross-Validation (CV) LASSO Regression*:
  - A CV LASSO regression that predicts the total number of bikes resulted in an optimal $\lambda = 12.216$ and a R^2_{OOS} of 0.83. 24 of 26 predictors were found to be statistically significant.

  - A CV LASSO regression that predicts the number of bikes rented by Capital Rideshare members resulted in an optimal $\lambda = 9.625$ and a R^2_{OOS} of 0.835. 23 of 26 predictors were found to be statistically significant.
  - A CV LASSO regression that predicts the number of bikes rented by non-members resulted in an optimal $\lambda = 3.731$ and a R^2_{OOS} of 0.716. 24 of 26 predictors were found to be statistically significant.

- *Clustering with K-Means*:
  - An initial k-means clustering of average temperature, humidity, and speed predictors at $k = 3$ resulted in fairly separable, nonoverlapping clusters. 233 observations were assigned to cluster 1, 215 to cluster 2, and 283 to cluster 3.
  - A subsequent cluster at $k = 5$ resulted in clusters with significantly more overlap. 213 observations were assigned to cluster 1, 156 to cluster 2, 117 to cluster 3, 91 to cluster 4, and 156 to cluster 5.
    * A k-means regression on the five clusters found all clusters to be statistically significant. However, the multiple R^2 is quite low (0.2634).
  - Selecting the optimal k using the elbow method resulted in $k = 5$ clusters being chosen.
  - Selecting the optimal k using the silhouette method had $k = 3$ with the lowest silhouette score of 0.200.

- *Principal Component Analysis (PCA)*:
  - PCA on Numeric variables only:
    * A screeplot of the principal analysis components on numeric variables only resulted in three principal components being chosen.
    * Conducting LASSO regression on the principal components resulted in 7 of 8 principal components being chosen (i.e. 7 of 8 principal components were statistically significant). The OOS R^2 for the LASSO regression was 0.507.
  - PCA on Numeric and Categorical variables:
    * A screeplot of the principal analysis components on a combination of numeric and categorical variables resulted in eight principal components being chosen.
    * Conducting LASSO regression on the principal components resulted in 23 out of 37 principal components being chosen. The OOS R^2 for the LASSO regression was 0.833.

- *Regression Trees*:
  - Regression Tree Model on Total Bike Rental Counts:
    * For the task of predicting total bike rental counts, the CV MSE was most reduced with a regression tree model of 7 splits.
    * The in-sample R^2 of the regression tree model of total bike rental counts was 0.8056.
  - Regression Tree Model on Bike Rental Counts for Registered Users:
    * For the task of predicting bike rental counts for registered users, the CV MSE was most reduced with a regression tree model of 10 splits.

    - The in-sample R^2 of the regression tree model of bike rental counts for registered users was 0.800.

- Regression Tree Model on Bike Rental Counts for Unregistered Users:
    - For the task of predicting bike rental counts for unregistered users, the CV MSE was most reduced with a regression tree model of 8 splits.
    - The in-sample R^2 of the regression tree model of bike rental counts for unregistered users was 0.804.

2 About the Dataset

The dataset was compiled by the UC Irvine Machine Learning Repository, and contains the counts of rental bikes between 2011-2012 in the Capital rideshare system.

The dataset contains are thirteen features, which are listed below:

Table 1: Feature Names in the Bikes Dataset

Feature Name
instant
dteday
season
yr
mnth
holiday
weekday
workingday
weathersit
temp
atemp
hum
windspeed
casual
registered
cnt

Specifically, each feature represents the following:

- **instant**: A record index of the rental transaction [nominal]
- **dteday**: The date of the rental transaction [categorical]
- **season**: A variable that contains values 1 to 4, each corresponding to a season (spring, summer, fall, winter, respectively) [categorical]
- **yr**: The year of the rental transaction; 0 represents 2011, 1 represents 2012 [categorical]
- **mnth**: The month of the rental tranaction; takes on values 0 to 12, with each value corresponding to a month in the year [categorical]
- **hr**: The hour of the rental transaction; takes on values 0 to 23, with each value corresponding to a hour in the 24-hour day [categorical]
- **holiday**: Records whether the day is a holiday or not; 0 represents non-holiday, 1 represents holiday [categorical]
- **weekday**: The day of the week; takes on values 0 to 7, with each value corresponding to a day of hte week [categorical]

- **weathersit**: Records the cloudiness of the particular day; 1 represents clear skies, 2 represents cloudy, 3 represents heavy rain, thunderstorm [categorical]
- **temp**: Records standardized temp of the hour in which the transaction occurred in degrees Celsius [numeric]
- **atemp**: Records the standardized "feels like" temperature in degrees Celsius [numeric]
- **hum**: Records the standardized humidity, with 100 being the maximum humidity [numeric]
- **windspeed**: Records the standardized windspeed (maximum windspeed ins 67) [numeric]
- **casual**: The total count of bike rentals from users unregistered to Capital Rideshare membership [numeric]
- **registered**: The total count of bike rentals from users registered to Capital Rideshare membership [numeric]
- **cnt**: The overall total of bike rentals logged in the Capital database [numeric]

3 Exploratory Analysis on Response

3.1 Distribution of Total Bike Rental Counts

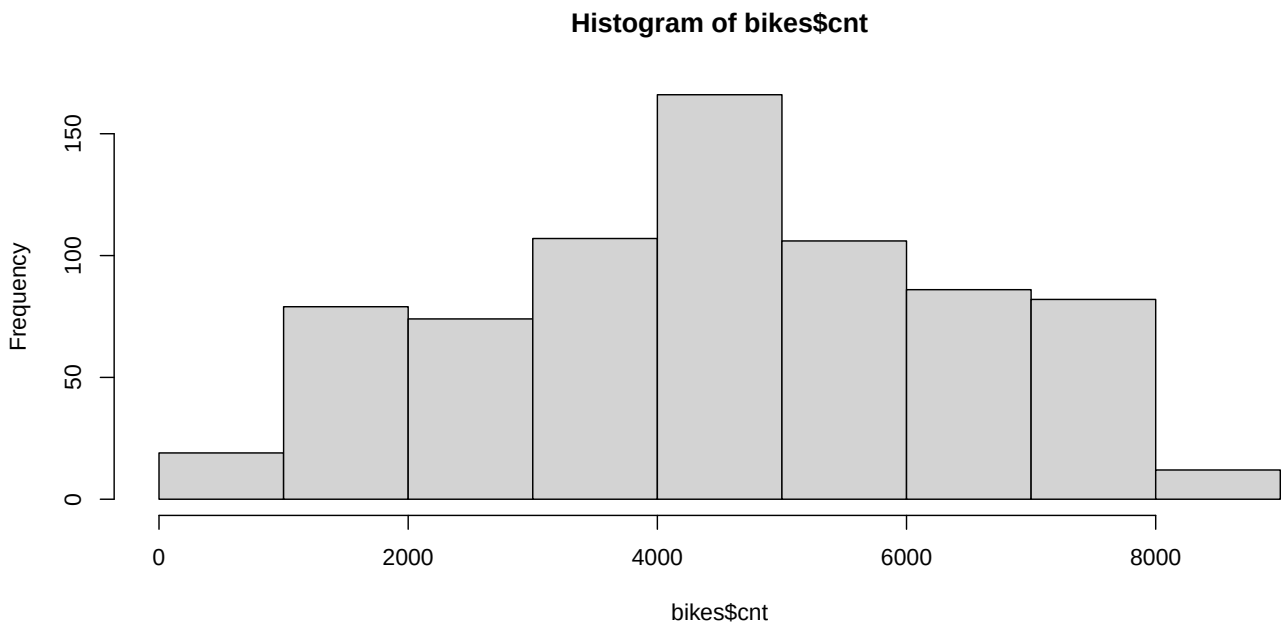


Table 2: Summary Statistics for Total Bikes Rented

Statistic	Value
Mean	4504.349
Median	4548.000
Interquartile Range (IQR)	2804.000
Standard Deviation	1937.211

From the histogram above, we can see the distribution of the overall amount bikes rented using the Capital rideshare system fro 2011 to 2012.

- The distribution is fairly unimodal and symmetric, as seen in the histogram. The mean (4504.35) is also fairly close to the median (4548.0), which also indicates that the distribution is symmetric.

- The IQR is quite large in comparison to the mean suggesting that the distribution of total bikes exhibits significant amounts of variation.

- We suspect that this variation likely stems from differences in the predictors of the dataset, which then cause a corresponding change in response. Thus, the primary objective of this analysis is to conduct statistical inference on how the response variable of interest y changes with the predictiors $x_i \dots x_n$.

3.2 Distributions of Bike Rental Counts for Registered and Unregistered Users

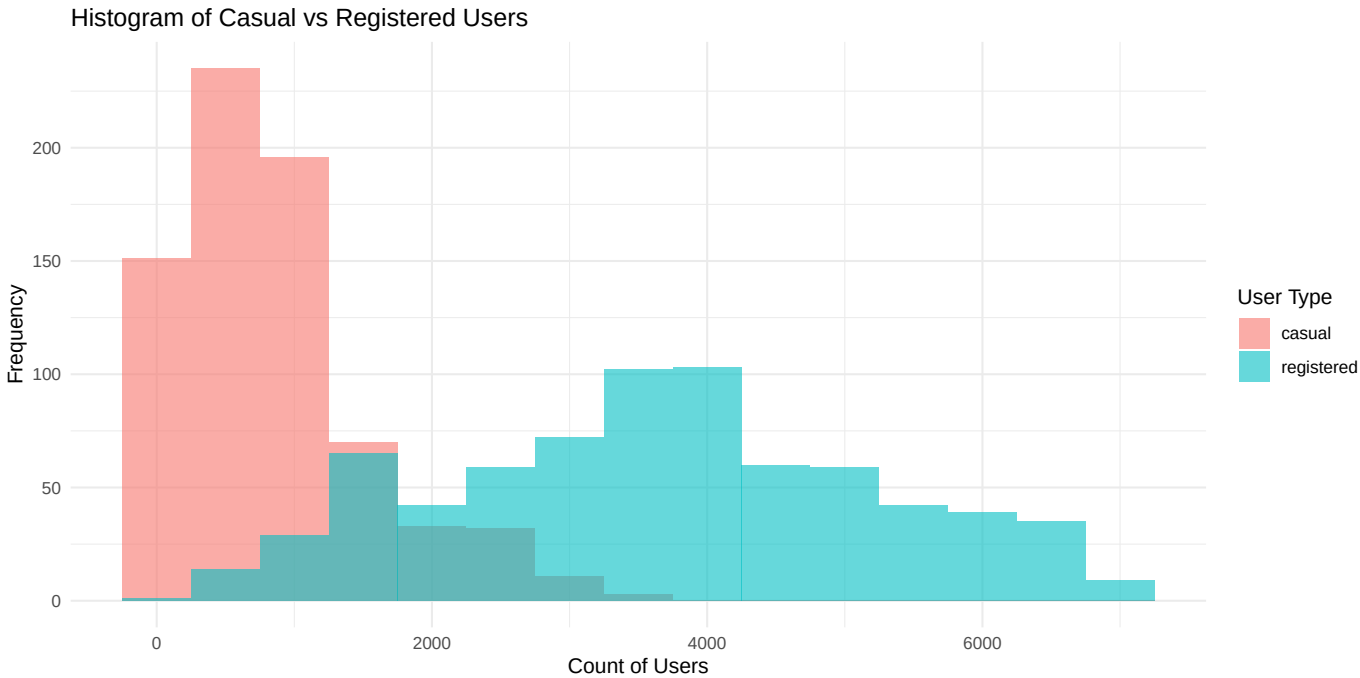
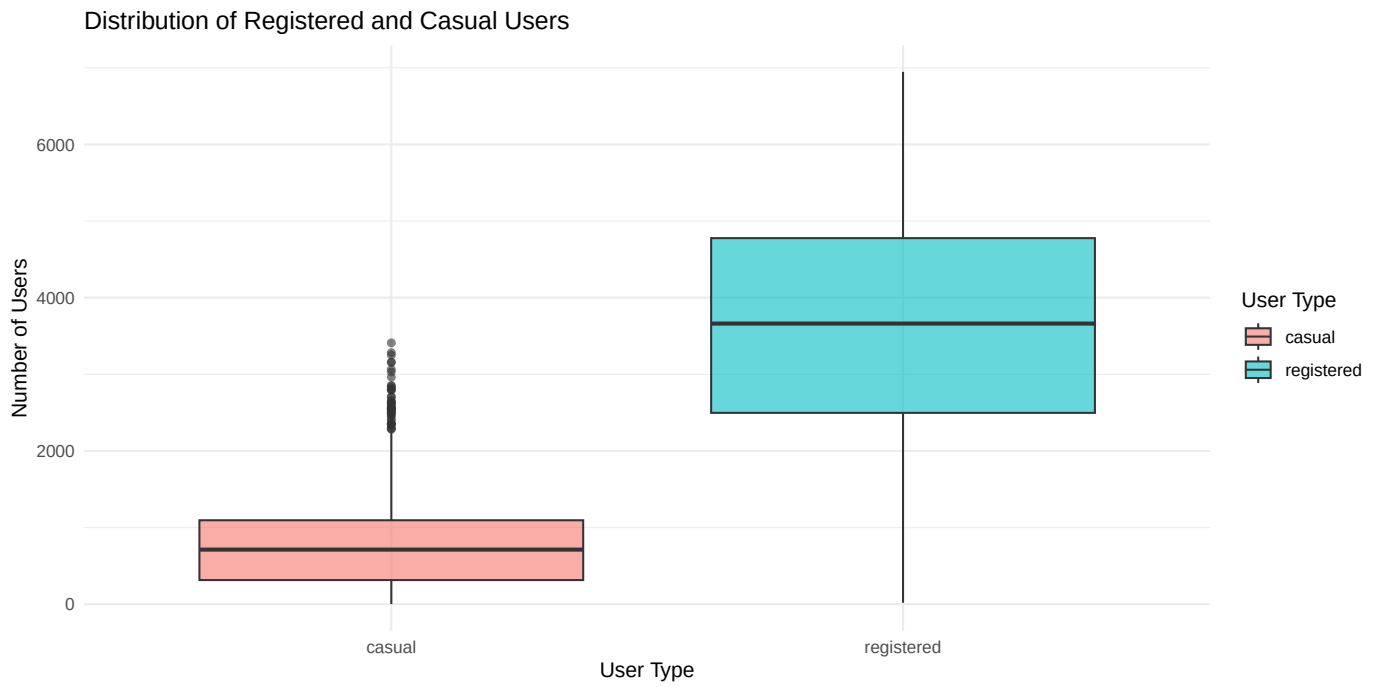


Table 3: Summary Statistics for Casual and Registered Users

Statistic	Casual Users	Registered Users
Mean	848.1765	3656.172
Median	713.0000	3662.000
Interquartile Range (IQR)	780.5000	2279.500
Standard Deviation	686.6225	1560.256

The histogram above compares the distribution of bike rented counts for users registered in the Capital bikeshare database (in blue) and for users not registered in the Capital bikeshare database (in red).

- From the plot, we can see that the distribution of the unregistered users is unimodal and skewed right. Th mean is also significantly greater than the median (848.18 and 713.0, respectively, which also indicates that the distribution is skewed right.
- On the other hand, the distribution of unregistered users is unimodal and roughly symmetric with a mean of 3656.17 and a median of 3662.0.
- Like the distribution of bike total rental counts, the IQRs of both distributions are quite large relative to the mean, suggesting that both distributions exhibit fairly high amounts of variation.

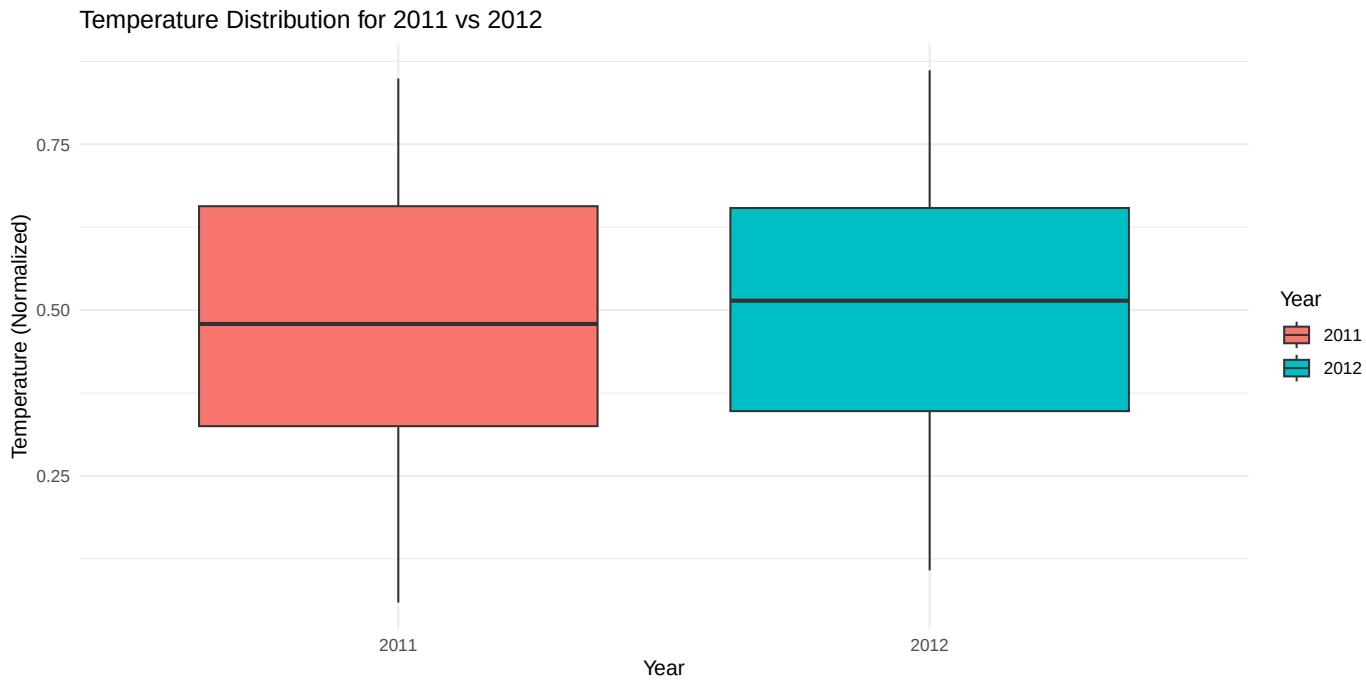


We can also visualize the bike rental count distributions with a side by side boxplot comparing casual to registered users.

- As consistent with the histogram, the center (median) of the bike rental counts for the unregistered users appears to be significantly less than that of the bike rental counts for for the registered users ($713 < 3662$).
- We also note that the distribution of bike rental counts for unregistered users has several outliers at the upper end of the distribution.

4 Exploratory Analysis on Predictors

4.1 Distribution of Temperatures of 2011 and 2012

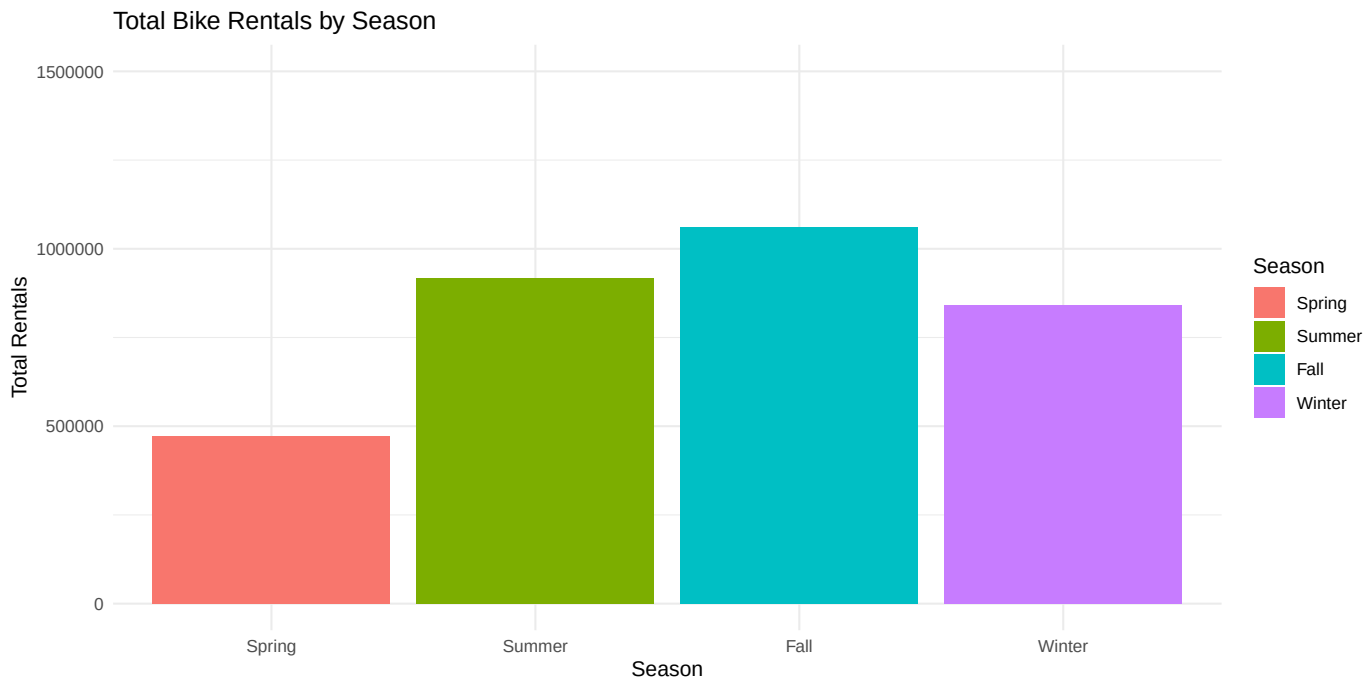


We are also interested in whether the distribution of temperatures between Year are substantially different or not (a significant difference in distributions can confound the causal effect between daily temperature measurements and bike rental counts).

- From the above boxplot, we can see that the distribution of temperatures for 2011 is indeed similar to that of 2012, with similar medians and similar quartile lengths.
- No significant outliers seem to be present in either distribution.

4.2 Distribution of Bike Rentals by Season

We would also like to measure how seasonal changes affect bikeshare counts. As a preliminary measurement, we first visualize the distribution between season and total bikes rented via a bar plot.



- The above distribution does indeed provide evidence that season does indeed have an effect on total bikeshare rentals, as there is a substantial difference in rentals from spring to summer. D
- Discrepancies between summer to fall and fall to winter are also evident from the distribution, but are less pronounced than that of spring to summer.

4.3 Distribution of Bike Rentals with respect to Holiday

We suspect that the total number of bike rentals on a particular day also change with whether that day is classified as a holiday or not.

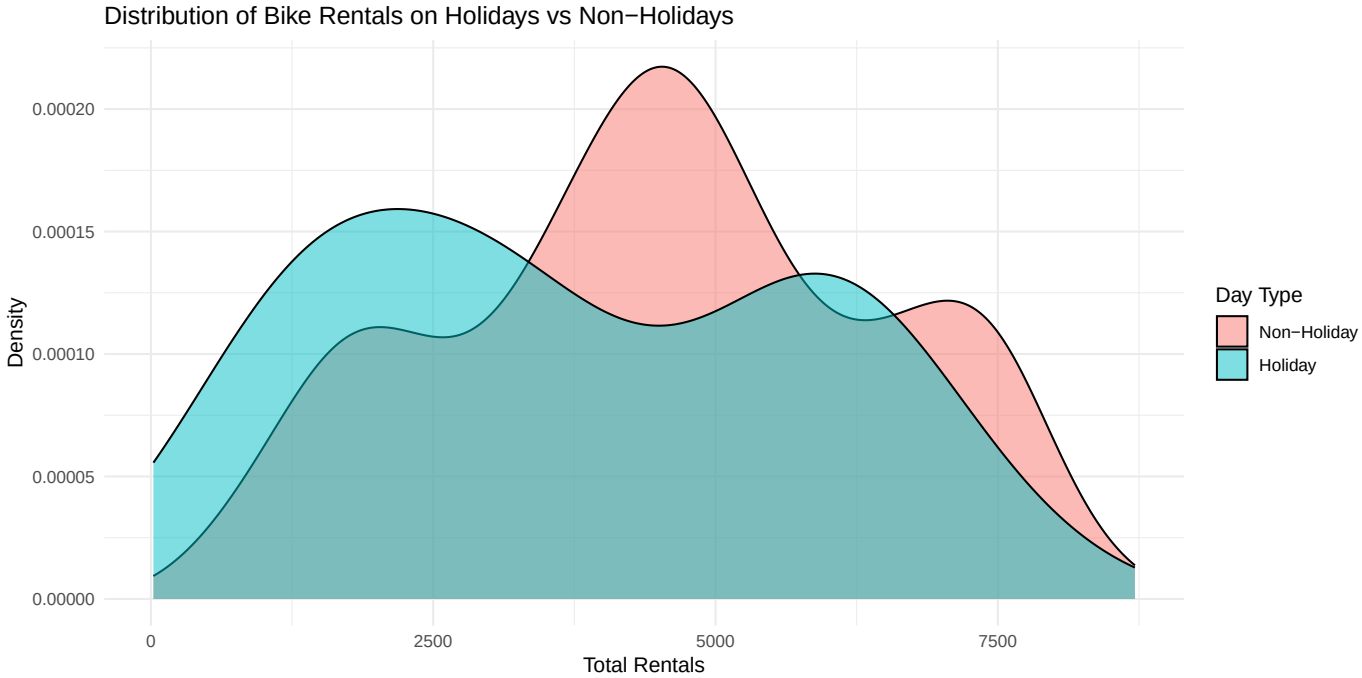
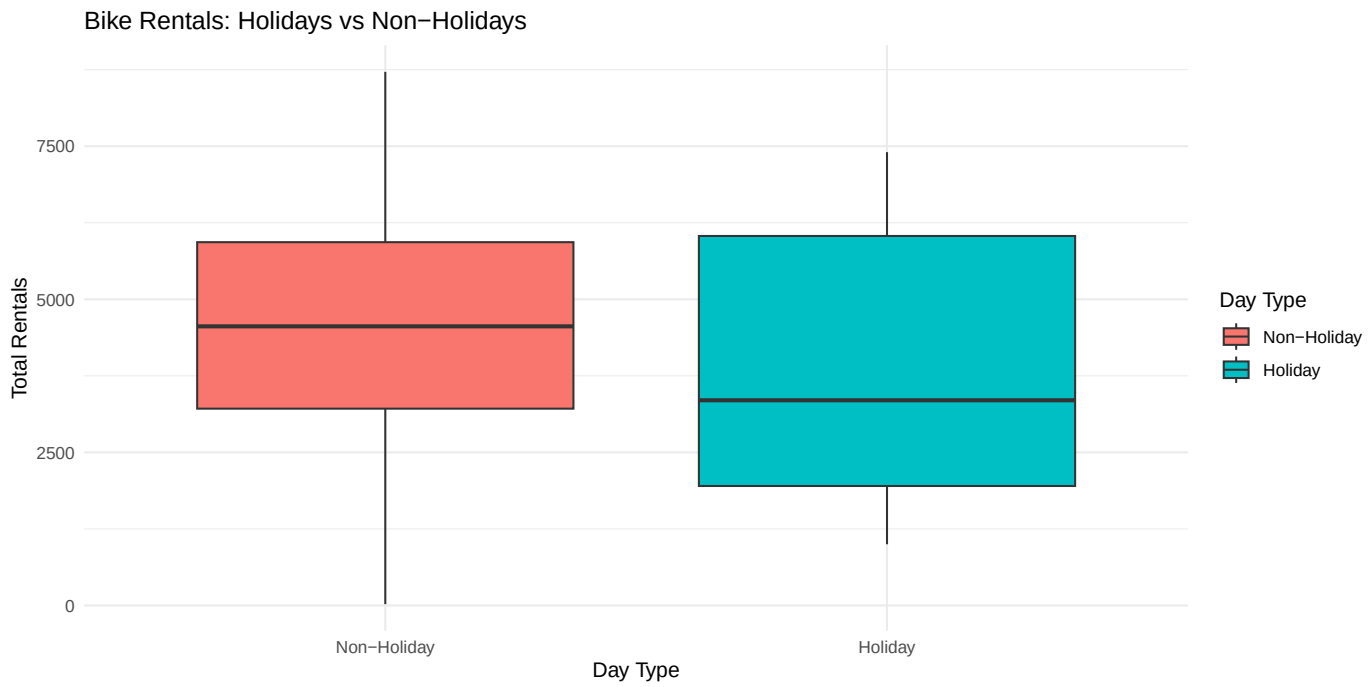


Table 4: Summary Statistics for Holiday vs. Non-Holiday Bike Rentals

Day Type	Mean	Median	IQR	Standard Deviation
Non-Holiday	4527.104	4558	2718.75	1929.014
Holiday	3735.000	3351	4083.00	2103.351

To visualize how the distribution of bike rentals depends on whether a particular day is a holiday or not, we plot overlapping density plots.

- We can see that the density plot of rentals on holiday is roughly bimodal with peaks around 1900 and 6250, while the density plot of rentals on non-holiday is unimodal with a peak at 4400.
- Interestingly, the peaks of the bike rental counts for the holiday distribution are aligned somewhat closely with the concavities of the bike rental counts for the non-holiday distribution, and vice versa.
- The mean of total bikes rented on non-holidays is actually greater than the mean of total bikes rented on holidays, which is perhaps somewhat counterintuitive. One potential reason is that the bikes in the Capital rideshare system are not typically used for recreational purposes, and are instead used to commute to work.



We can again visualize the distribution of the bike rental counts with a side-by-side boxplot.

- As consistent with the density plot, the median of the bike rental counts for non-holidays is 4558, and the median of of the bike rental counts for holidays is 3351.
- There also seems to be a significantly greater amount of variation in the distribution of the bike rental counts for non-holidays, as evident by the long tails of the boxplot.
- On the other hand, the distribution of counts for holidays appears to be somewhat more concentrated around the median, as evident by the shorter tails of the boxplot. It is important to note that the interquartile range of the distribution of counts for holidays is indeed greater than that of non-holidays.

5 Prediction using Naive Regression

5.1 Motivation

In this section, we conduct a more in-depth analysis on how the predictors present (e.g. humidity, temperature, season) in the bikes dataset affect the outcome of interest (the number of rideshare bikes rented out on a particular day). To do this, we will use several tools that aim to determine the statistical significance of a given predictor on the outcome, beginning with naive regression.

5.2 Naive Regression

5.2.1 Predictors used in Regression Analysis

In naive regression, we typically would regress count onto all covariates, and directly report the estimated coefficients and their corresponding P-values. However, there are firstly some covariates that are dependent on one another that will skew the fit significantly if not accounted for.

Firstly, we have the following relationship between bike rental count and the registration status of a particular user.

Define the total bike rental count as T, the total bike rentals from registered users as R, and the total bike rentals from unregistered users as U. Then,

$$T = R + U$$

Thus, as there is an inherent dependency between the total bike rental count (and the total bike rental count from registered and unregistered users, we will typically choose to only treat one of the three variables (total, registered, unregistered) as the response variable of interest, and omit the other two variables from the model entirely.

In the following analyses, we also choose to remove the "instant" variable from our constructed models, as it is simply an index for the particular day. We also choose to remove the "dteday" variable from our constructed regressions, as it would be treated as a factor with $365 \cdot 2 = 730$ levels, each corresponding to a particular day in the two-year study. We note that such a variable is not particularly helpful to our analysis, as we are focused on the overall statistical significance of a given predictor over the entire two-year period, not just on a particular day. We instead find the "Year" column to be a better predictor that can capture overall trends in bike rental counts, and use that in our model instead.

5.2.2 Converting Variables to Factors

In the dataset, we also have several predictors that are categorical rather than numeric. We typically handle these predictors in three ways:

1. We can simply put the categorical predictors in the model (used for Naive Regression).
2. We can choose to ignore the categorical predictors entirely, and focus only on the numeric predictors. This approach is typically avoided, and in the section regarding Principal Component Analysis (PCA) we show why by simply ignoring categorical variables is not optimal by comparing the accuracy of a model regressed only on principal components constructed from numeric

variables versus the accuracy of a model regressed on principal components constructed from numeric and categorical variables.

3. We can treat the categorical variables as numeric by assigning each unique category a number (i.e. we treat the categorical variable as a factor).

Table 5: Number of Levels in Factor Variables

Variable	Number of Levels
season	4
yr	2
mnth	12
holiday	2
weekday	7
workingday	2
weathersit	3

5.2.3 Naive Linear Regression on Total Bike Rental Counts

We first regress the total counts of bike rentals onto the full set of covariates, excluding "instant", "dteday", "casual", and "registered".

Table 6: Standard Regression Coefficients with all Covariates among All Users

Coefficient	Estimate	Std. Error	P-Value
(Intercept)	1.4690e+03	2.4022e+02	1.5813e-09
season	5.0978e+02	5.4757e+01	1.5239e-19
yr	2.0407e+03	6.5185e+01	2.1123e-136
mnth	-3.8980e+01	1.7079e+01	2.2763e-02
holiday	-5.1899e+02	2.0104e+02	1.0033e-02
weekday	6.9062e+01	1.6299e+01	2.5576e-05
workingday	1.2036e+02	7.2007e+01	9.5067e-02
weathersit	-6.1099e+02	7.8363e+01	2.2312e-14
temp	2.0289e+03	1.4037e+03	1.4877e-01
atemp	3.5733e+03	1.5894e+03	2.4865e-02
hum	-1.0189e+03	3.1400e+02	1.2296e-03
windspeed	-2.5576e+03	4.5628e+02	2.9641e-08

The coefficients of each predictor along with a corresponding P-value are displayed in Table 1. We can see that for a naive linear regression onto the selected set of covariates, most coefficients had a statisticallly significant two-tailed P-value. The significance of these coefficients are also supported by a fairly large in-sample $R^2 = 0.800$. However, we also find that some predictors, such as month ("mth"), have a fairly small coefficient ($\hat{\beta}_{mth} = -3.898$). Other predictors, such as the "feels like" temperature ("atemp"), have coefficients greater in magnitude but also larger standard errors ($SE_{atemp} = 3.5733 \cdot 10^3$).

5.2.4 Naive Regression on Counts Among Registered Users

Table 7: Standard Regression Coefficients with all Covariates among Registered Users

Coefficient	Estimate	Std. Error	P-Value
(Intercept)	7.6821e+02	1.8555e+02	3.8825e-05
season	4.4795e+02	4.2296e+01	1.8648e-24
yr	1.7540e+03	5.0351e+01	1.6028e-156
mnth	-2.3235e+01	1.3192e+01	7.8618e-02
holiday	-2.4478e+02	1.5529e+02	1.1540e-01
weekday	4.2699e+01	1.2590e+01	7.3303e-04
workingday	9.4861e+02	5.5620e+01	5.0377e-55
weathersit	-4.9794e+02	6.0530e+01	9.0391e-16
temp	8.3407e+02	1.0842e+03	4.4198e-01
atemp	2.6784e+03	1.2277e+03	2.9456e-02
hum	-6.2563e+02	2.4254e+02	1.0091e-02
windspeed	-1.6955e+03	3.5244e+02	1.8325e-06

Now, we repeat the naive linear regression model, but this time on the bike rental counts of registered users only (hence removing "cnt" and "casual" from the model). Table 2 shows the results of the regression. We note several substantial differences in the coefficients listed in Table 2 and the coefficients from the above section (a naive regression model on the bike rental counts of all users:

- $\hat{\beta}_{holiday}$ increases from -518.99 to -244.78 (2.120 times difference).
- $\hat{\beta}_{workingday}$ increases from 120.36 to 948.61 (7.881 times difference).
- $\hat{\beta}_{temp}$ increases decreases from 2028.9 to 834.1 (0.410 times difference).

The in-sample R^2 is 0.816, which is fairly close to the in-sample R^2 for total bike rentals. We conclude that the accuracy of predicting total bike rentals and bike rentals among members only from covariates using naive regression is roughly similar.

5.2.5 Naive Regression on Counts Among Unregistered Users

Table 8: Standard Regression Coefficients with all Covariates among Unregistered Users

Coefficient	Estimate	Std. Error	P-Value
(Intercept)	7.0079e+02	1.0636e+02	8.5612e-11
season	6.1824e+01	2.4244e+01	1.0977e-02
yr	2.8668e+02	2.8861e+01	7.0933e-22
mnth	-1.5744e+01	7.5619e+00	3.7690e-02
holiday	-2.7421e+02	8.9012e+01	2.1444e-03
weekday	2.6364e+01	7.2165e+00	2.7778e-04
workingday	-8.2825e+02	3.1882e+01	1.8764e-105
weathersit	-1.1305e+02	3.4696e+01	1.1736e-03
temp	1.1948e+03	6.2149e+02	5.4931e-02
atemp	8.9486e+02	7.0371e+02	2.0392e-01
hum	-3.9323e+02	1.3902e+02	4.8067e-03
windspeed	-8.6205e+02	2.0202e+02	2.2448e-05

Lastly, we repeat the naive linear regression for unregistered users only (thus removing "cnt" and "registered" from the model). Table 3 shows the results of the regression.

Comparing this model to the model from ii(e), i.e. comparing the model where the response variable is the bike rental count of registered users versus the bike rental count of unregistered users, we observe the following significant changes in predictors:

- $\hat{\beta}_{season}$ decreases from 447.95 to 61.82 (0.138 times difference)
- $\hat{\beta}_{yr}$ decreases from 1754.0 to 286.68 (0.163 times difference)
- $\hat{\beta}_{weathersit}$ increases from -497.94 to -113.05 (4.405 times difference)
- $\hat{\beta}_{atemp}$ decreases from 2678.4 to 894.86 (0.334 times difference)

Comparing this model to the model from ii(c), i.e. comparing the model where the response variable is the bike rental count of all users versus the bike rental count of unregistered users, we observe the following significant changes in predictors:

- $\hat{\beta}_{season}$ decreases from 1469.0 to 700.79 (0.477 times difference)
- $\hat{\beta}_{yr}$ decreases from 509.7 to 61.82 (0.121 times difference)
- $\hat{\beta}_{workingday}$ decreases from 120.36 to -828.25 (-6.88 times difference)
- $\hat{\beta}_{weathersit}$ decreases from -610.99 to -113.05 (0.185 times difference)
- $\hat{\beta}_{atemp}$ decreases from 3573.3 to 894.86 (0.250 times difference)

We can see that both sets of model comparisons resulted in significant differences to a similar set of predictors (e.g. $\hat{\beta}_{atemp}$ had a similar times difference for both comparisons).

The in-sample R^2 for the task of predicting bike rentals among unregistered users is 0.688. We conclude that the accuracy of predicting bike rentals among non-members only from covariates using naive regression is less accurate than the other two prediction tasks.

6 LASSO Regression on Counts

6.1 Motivation

In previous sections, we have identified potential issues with the dataset that may affect the accuracy of naive regression. For example, dependencies may exist between predictors that we use to fit the naive regression model, which violates a central assumption of naive regression that the predictors are independent.

Furthermore, by adding a L1 regularization term, LASSO effectively prunes the model by shrinking coefficients to 0, thus removing any covariates whose effects on the response variable may not be significant. The regularization parameter λ effectively controls the rate at which coefficients are being shrunk to 0.

It is important note that in naive regression we can simply put categorical variables as predictors in the model without altering them. However, LASSO requires all variables to be numeric, so we convert all categorical variables to numeric predictors by transforming them to factors using the procedure outlined in ii(b).

6.2 Cross-Validation (CV) LASSO of Total Bike Rental Counts

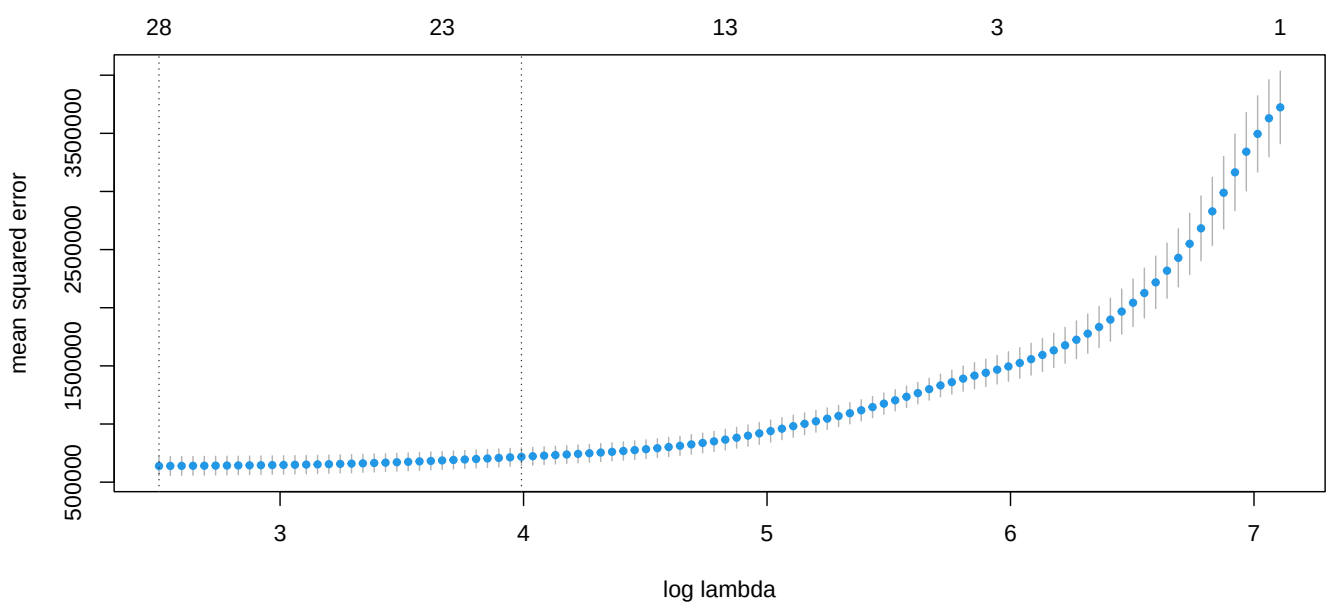


Table 9: CV LASSO Regression Coefficients

	Predictor	Coefficient Value
intercept	intercept	1515.2583
season2	season2	956.7908
season3	season3	666.2243

	Predictor	Coefficient Value
season4	season4	1306.0773
yr1	yr1	1983.8220
mnth2	mnth2	0.0000
mnth3	mnth3	243.1301
mnth4	mnth4	46.1443
mnth5	mnth5	251.8103
mnth6	mnth6	53.0986
mnth7	mnth7	-241.8188
mnth8	mnth8	123.7359
mnth9	mnth9	769.2565
mnth10	mnth10	448.2860
mnth11	mnth11	-16.6821
mnth12	mnth12	-60.3484
holiday1	holiday1	-301.5790
weekday1	weekday1	-108.6607
weekday2	weekday2	-13.1890
weekday3	weekday3	0.0000
weekday4	weekday4	5.2734
weekday5	weekday5	59.7114
weekday6	weekday6	348.9345
workingday1	workingday1	282.0704
weathersit2	weathersit2	-419.6805
weathersit3	weathersit3	-1863.2372
temp	temp	3665.9523
atemp	atemp	1666.2575
hum	hum	-1571.2863
windspeed	windspeed	-2729.0044

The above plot shows MSE with respect to $\ln(\lambda)$. We can see that the $\ln(\lambda)$ that corresponds to the lowest Mean Squared Error (MSE) is around 2.50 and 3.50. Indeed, the optimal lambda that minimizes MSE is at 12.216, which corresponds to a $\ln(\lambda)$ value of 2.503.

The table shows the resulting coefficients from CV LASSO with $\lambda = 12.216$. We can see that 24 out of the 26 original predictors were selected by CV LASSO to be statistically significant, with only "mnth2" and "weekday3" being shrunk to zero.

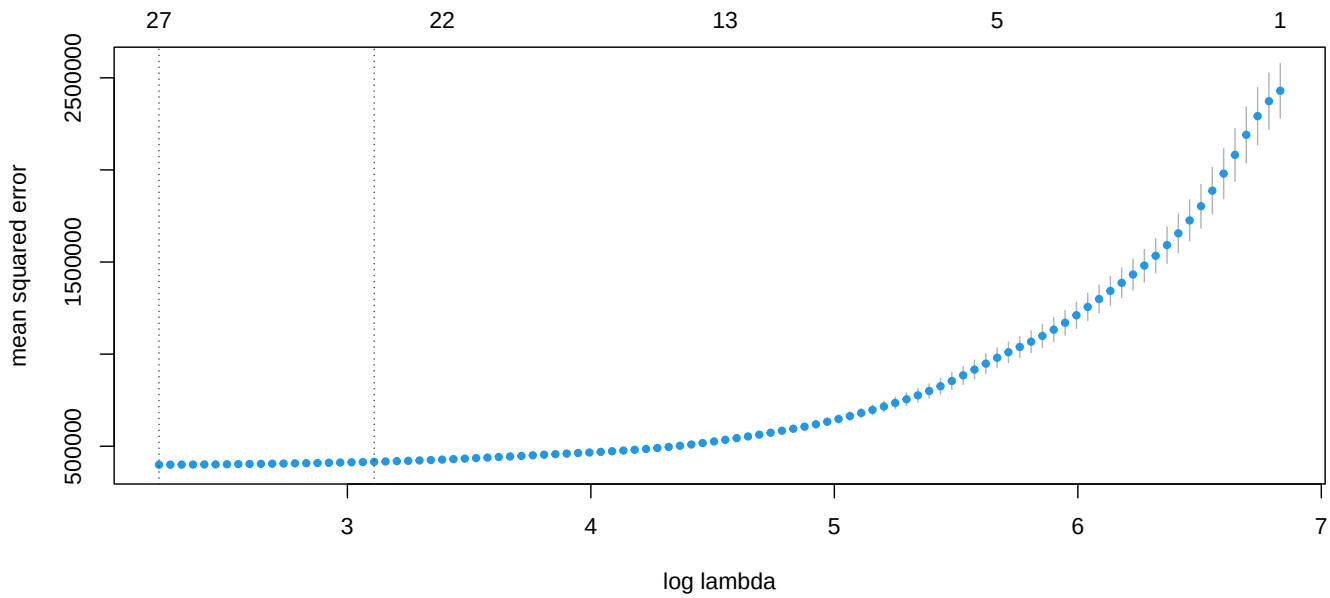
As our LASSO model was trained using cross-validation (meaning that we reserve a fold for testing), we can calculate the out-of-sample (OOS) with the following expression:

$$R^2_{OOS} = 1 - \frac{MSE_{OOS}}{\text{Var}(Y)}$$

where MSE_{OOS} is the MSE of the LASSO model on the holdout fold, and $Var(Y)$ is the variance of the response variable.

The R^2_{OOS} of this model (LASSO regression on total bike rental counts) is 0.8303.

6.3 CV LASSO of Bike Rental Counts among Registered Users




```
Best Lambda Selected:  9.265451
```

Table 10: CV LASSO Regression Coefficients

	Predictor	Coefficient Value
intercept	intercept	801.5936
season2	season2	635.6126
season3	season3	584.5903
season4	season4	1301.6921
yr1	yr1	1709.3223
mnth2	mnth2	0.0000
mnth3	mnth3	81.4297
mnth4	mnth4	0.0000
mnth5	mnth5	196.7167
mnth6	mnth6	141.6915
mnth7	mnth7	-179.7185
mnth8	mnth8	117.8927
mnth9	mnth9	551.3964
mnth10	mnth10	114.6218
mnth11	mnth11	-185.7704
mnth12	mnth12	-70.8285
holiday1	holiday1	-62.9833
weekday1	weekday1	-110.5492
weekday2	weekday2	0.0000
weekday3	weekday3	72.3508

	Predictor	Coefficient Value
weekday4	weekday4	71.1436
weekday5	weekday5	-4.1079
weekday6	weekday6	222.5300
workingday1	workingday1	1034.3381
weathersit2	weathersit2	-329.7949
weathersit3	weathersit3	-1565.6816
temp	temp	1756.1393
atemp	atemp	1675.9479
hum	hum	-1002.2937
windspeed	windspeed	-1643.4303

From the above plot, we can see that the lambda that minimizes MSE for the CV LASSO model predicting bike rental counts among registered users is around 2.3 and 3.6. The optimal λ is indeed close to our estimate ($\ln(\lambda) = 2.226 \rightarrow \lambda = 9.625$).

From Table 6, we can see that similar to the CV LASSO model in ii(b), 23 out of 26 predictors were deemed statistically significant. The three predictors that were deemed statistically insignificant (i.e. their coefficients converged to 0) were "mnth2", "mnth4", and "weekday2".

The OOS R^2 of the CV LASSO model is 0.83517. There is not much difference between the overall accuracy of predicting total bike rental count versus bike rental count of registered users for the Capital bikeshare system.

6.4 CV LASSO of Bike Rental Counts among Unregistered Users

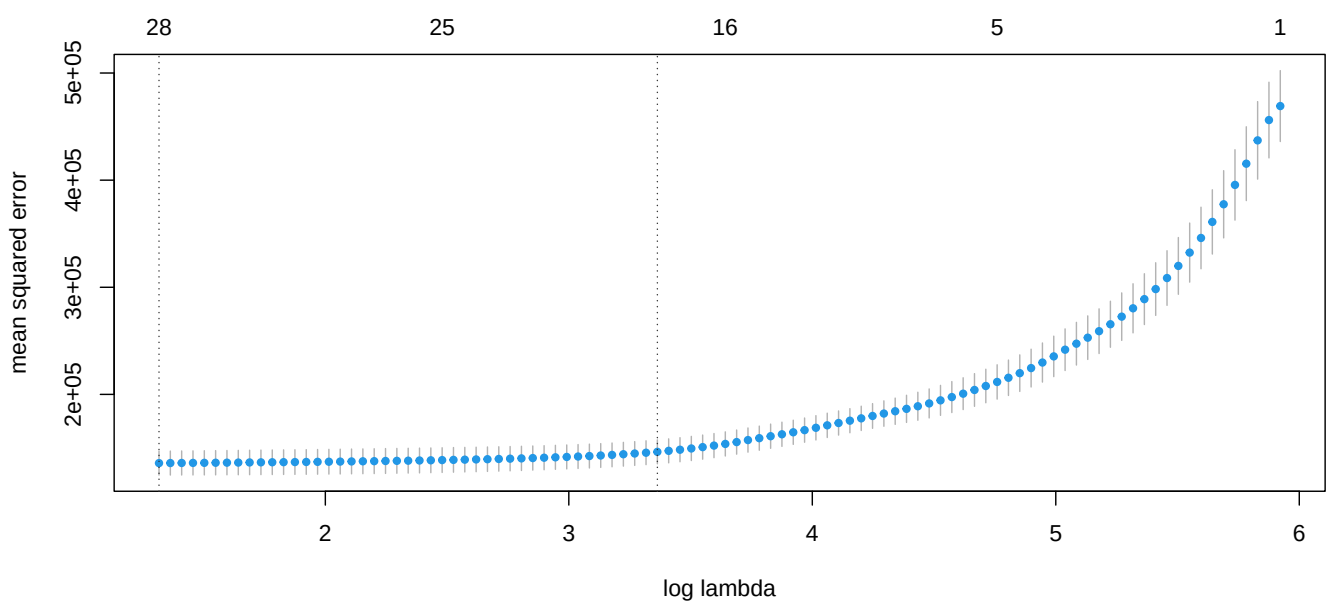


Table 11: CV LASSO Regression Coefficients

	Predictor	Coefficient Value
intercept	intercept	702.6506
season2	season2	228.6786
season3	season3	33.2198
season4	season4	36.2707
yr1	yr1	276.3356
mnth2	mnth2	-65.2519
mnth3	mnth3	169.4454
mnth4	mnth4	141.6891
mnth5	mnth5	137.3523
mnth6	mnth6	0.0000
mnth7	mnth7	-1.5494
mnth8	mnth8	56.8545
mnth9	mnth9	234.2445
mnth10	mnth10	280.9990
mnth11	mnth11	86.8137
mnth12	mnth12	-38.9011
holiday1	holiday1	-208.0908
weekday1	weekday1	0.0000
weekday2	weekday2	-43.8749
weekday3	weekday3	-48.2008
weekday4	weekday4	-37.2389
weekday5	weekday5	118.9224
weekday6	weekday6	147.4928
workingday1	workingday1	-738.8637
weathersit2	weathersit2	-96.9039
weathersit3	weathersit3	-297.2207
temp	temp	1669.6153
atemp	atemp	166.5761
hum	hum	-518.2626
windspeed	windspeed	-1048.2618

From the above plot, we can see that the lambda that for the CV LASSO model predicting bike rental counts among unregistered users, the optimal $\ln(\lambda)$ is around 1.3 and 3.1. Indeed, the optimal lambda is $\ln(\lambda) = 1.3166 \rightarrow \lambda = 3.731$.

The OOS R^2 is 0.7155, implying that the CV LASSO to predict bike rentals for unregistered users is lower than the CV LASSO models to predict bike rentals among registered users and total bike rentals.

- This is perhaps expected, as unregistered users may only rent a bike to suit a specific purpose (unlike the registered users who may regularly rent a bike, making it easier to draw predictions).

6.5 Treatment Effect of Season on Registered Users using 2-Stage LASSO

6.5.1 Motivation for 2-Stage LASSO

We discovered in Section 2.3 that the distribution of bike rentals by season was not uniform; that is, there was evidence that season and total bike share rentals were associated. However, this relationship is not causal - from a bar plot alone, we are unable to definitively say that season causes a change in total bike share rentals.

2-Stage LASSO (also known as Causal LASSO) allows us to make such cause-and-effect statements by defining a treatment variable d and then isolating the casual effect of the treatment variable, ν.

To isolate the effects of d, we have to run a two-stage LASSO:

- 1st Stage: Predict $\hat{d}$ by regressing d (Season) on the rest of the covariates x. This controls for any confounding effects the covariates may have on the treatment effect.
- 2nd Stage: Regress response y on d, $\hat{d}$, and x. Note that $\hat{d}$ is unpenalized to ensure that it is accounted for in the LASSO model.

6.5.2 Results of 2-Stage LASSO

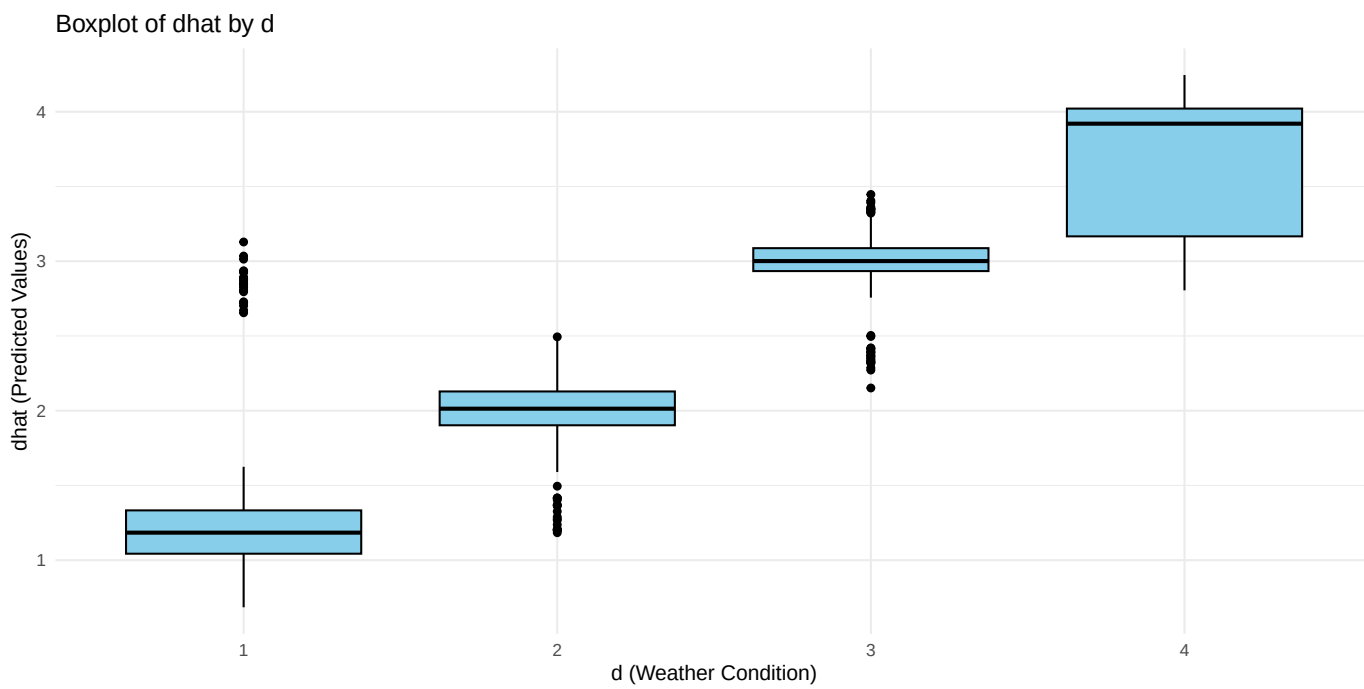


Table 12: 2-Stage LASSO Results

Metric	Value
In Sample R^2	0.8153825
OOS R^2	0.7983641
Coefficient of d	471.8045868

- From the boxplots comparing d to $\hat{d}$, we can see that as d (season) increases by one level, the median of the distribution of $\hat{d}$ also increases by one level, accordingly.
- However, for the corresponding $\hat{d}$ distribution at $d \in \{1, 2, 3\}$, outliers for $\hat{d}$ are present. For example, when $d = 1$, there are some $\hat{d}$ predictions that are 3.
- Both the in-sample and OOS R^2 are quite high (0.815 and 0.803, respectively). This provides evidence that our 2-Stage LASSO provided an accurate treatment effect estimation for d (Season).
- Interpretation for treatment effect d: When the season increases by 1 (Spring $\rightarrow$ Summer, Summer $\rightarrow$ Fall, Fall $\rightarrow$ Winter), the predicted number of bikes rented via the Capital bikeshare system increases by 471.80 bikes.

7 Clustering

7.1 Motivation

We have now examined the strength of each covariate to predict one of the three responses: total bike rentals, total bike rentals from registered users, total bike rentals from unregistered users. We have accounted for dependence between predictors (multicollinearity) by implementing a LASSO model. However, there may be inherent associations between observations themselves that can aid in performance.

Clustering, or grouping observations together based on similarities, can help us detect such inherent associations. The primary clustering algorithm we will be using in this analysis is the K-Means algorithm. The K-Means algorithm begins by initializing centroids (means), and then classifying each point to the closest centroid, thus forming a "cluster" of points centered around the centroid. Then, the centroid is updated to be the mean of the current set of points associated with it. This process repeats itself until the centroids converge at a particular point.

In particular, we suspect that a dependence exists between observations on how "comfortable" one feels outside (i.e. the predictors "atemp", "hum", "windspeed"). Thus, we aim to analyze the results of K-means on these three predictors.

7.2 K-Means with K = 3 Clusters

To begin, we selected a small number of clusters $k = 3$ (we will establish procedures for finding optimal k further into analysis).

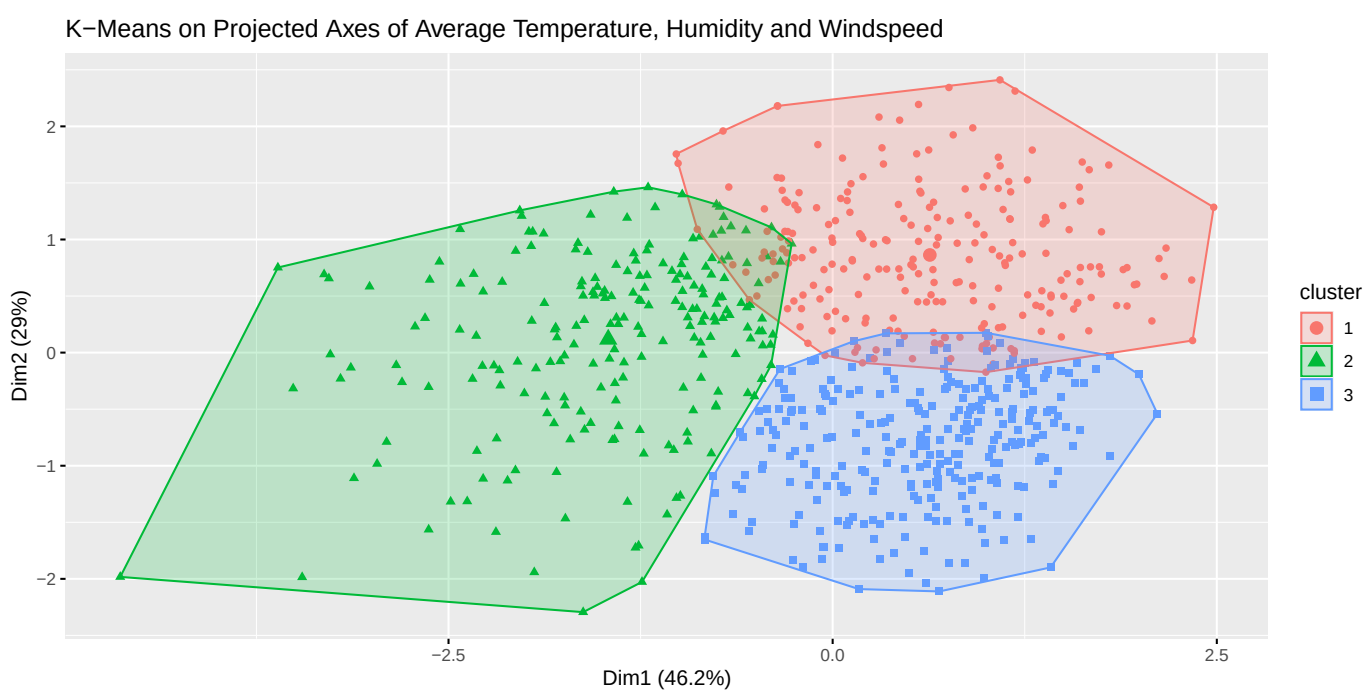


Table 13: Cluster Size Summary

Cluster	Number of Points
Cluster 1	233
Cluster 2	215
Cluster 3	283

The above plot shows the results of k-means with $k = 3$ clusters on a projected 2-D axes of Average Temperature, Humidity, and Wind Speed. We observe the following:

- The K-means algorithm seems to have identified three primary centers of interest- lower left, lower right, and top right.
- From the table, we can see that the distribution of observations to cluster 1 and cluster 2 is roughly even. Cluster 3 has more observations than the other two clusters.
- There does not seem to be significant amounts of overlap between the clusters, suggesting that the data is indeed separable. Perhaps there is a possible dependence between observations based on each cluster.

7.2.1 K-Means Regression on Total Bike Rentals with K = 3:

```
Call:
lm(formula = cnt ~ cluster, data = df)

Residuals:
    Min      1Q  Median      3Q     Max 
-4723.9 -1301.4   -15.9  1270.6  4350.3 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   3757.9      106.2  35.390   <2e-16 ***
cluster2      -201.2      153.3  -1.313     0.19    
cluster3      2081.1      143.4  14.514   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1621 on 728 degrees of freedom
Multiple R-squared:  0.3019,    Adjusted R-squared:    0.3 
F-statistic: 157.4 on 2 and 728 DF,  p-value: < 2.2e-16
```

We can see that while cluster 2 is not statistically significant, cluster 3 is. Perhaps further increasing the number of clusters allows us to determine a trend in the weather patterns and how they predict the total number of bike rentals.

7.3 K-Means with K = 5 Clusters:

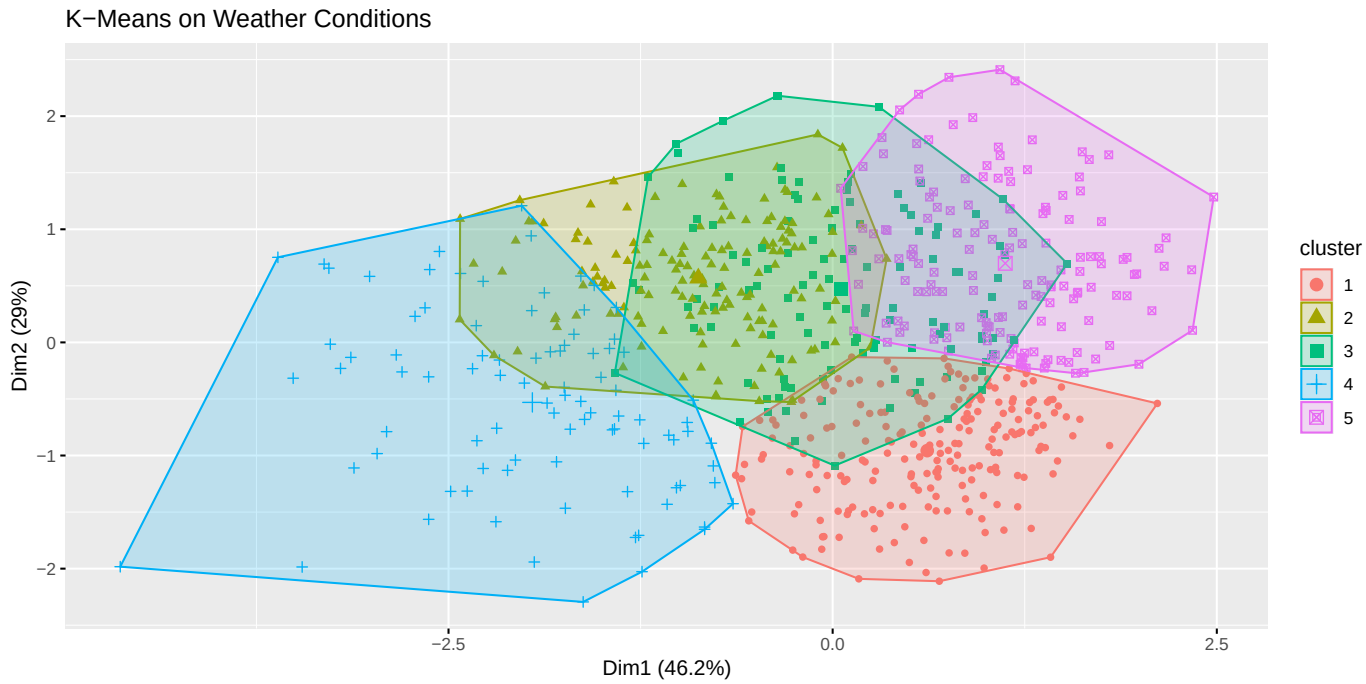


Table 14: Cluster Size Summary

Cluster	Number of Points
Cluster 1	213
Cluster 2	154
Cluster 3	117
Cluster 4	91
Cluster 5	156

We can see that choosing a cluster of 5 seems to cause heavy overlap, especially for clusters 2 and 3. However, this overlap may in part be because "Dim1" and "Dim2" do not represent the entire feature space but rather are projections (principal components) of the 3-D feature space onto a 2-D plane; due to this projection, we are losing some information about the clusters.

The distribution of points between $k = 5$ clusters also seems to be less homogenous than the distribution of points with $k = 3$ clusters. For example, cluster 1 now has 213 observations, which is more than two times greater than the number observations in cluster 4 (which has 91 observations).

7.3.1 K-Means Regression on Total Bike Rentals with K = 5:

```
Call:
lm(formula = cnt ~ cluster, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-3991.9 -1290.0   -34.5  1255.2  4923.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5881.0      114.2  51.481  < 2e-16 ***
cluster2     -2443.5      176.4 -13.856  < 2e-16 ***
cluster3     -2409.2      191.9 -12.557  < 2e-16 ***
cluster4     -1758.4      208.8  -8.421  < 2e-16 ***
cluster5     -1206.2      175.7  -6.865 1.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1667 on 726 degrees of freedom
Multiple R-squared:  0.2634,    Adjusted R-squared:  0.2593
F-statistic: 64.89 on 4 and 726 DF,  p-value: < 2.2e-16
```

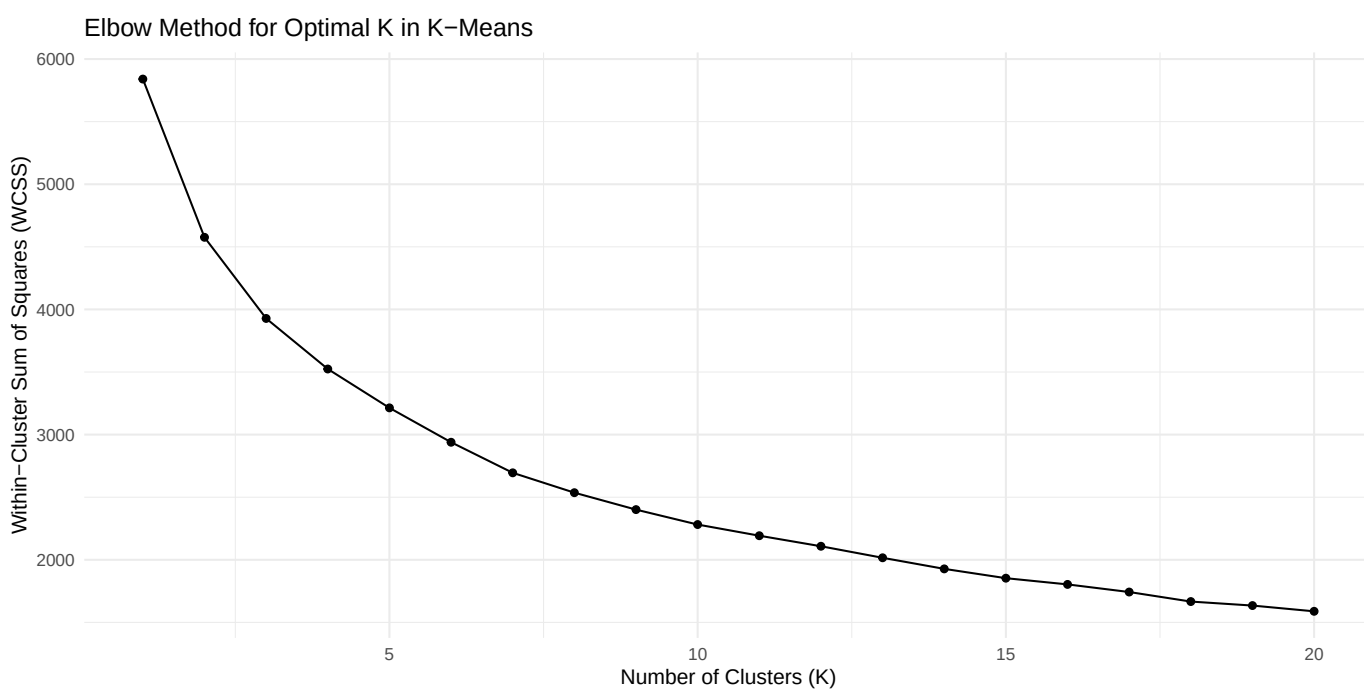
Even though the plot above showed significant overlap between clusters 2 and 3, all P-values are now statistically significant, suggesting that each cluster is indeed important at predicting the total bike rentals.

- Thus, it is likely that clusters 2 and 3 do not overlap significantly in the original 3-D feature space, and only overlap when projected onto the 2-D subspace.

However, the multiple R^2 is quite low (0.2634), suggesting that while our predictors are individually significant, they do not explain much of the total variation in total bike rentals.

7.4 Choosing Optimal Number of Clusters using Elbow Method

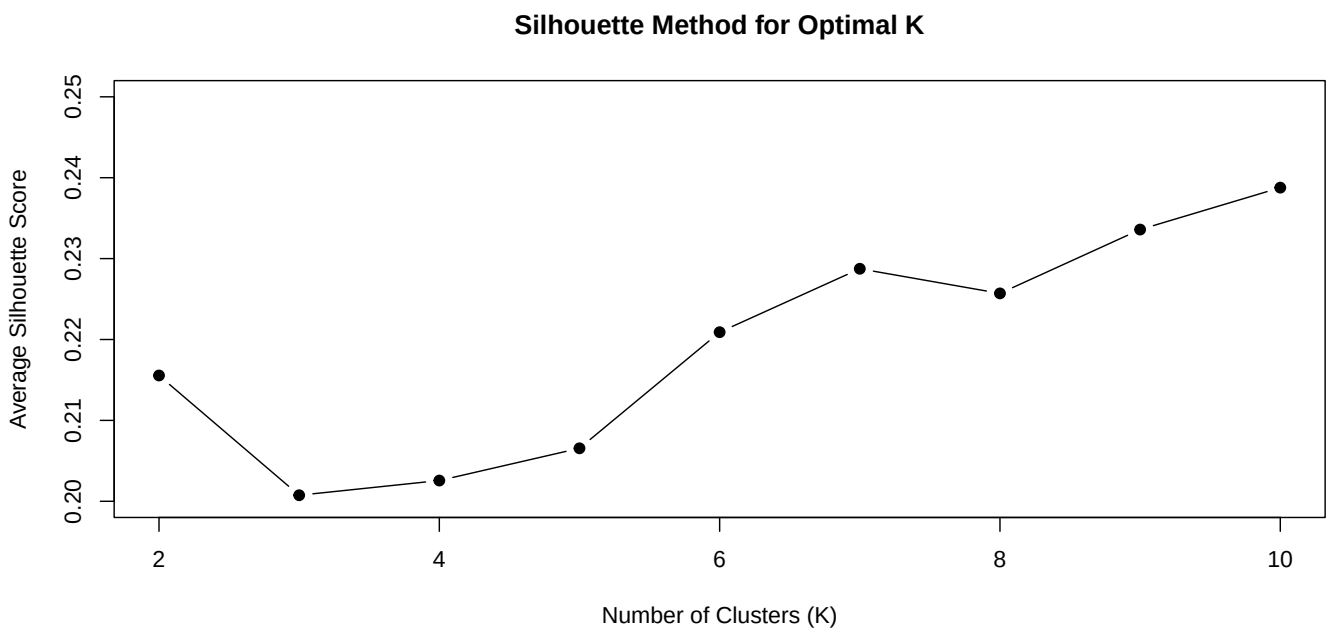
We have seen that the selecting the number of clusters k greatly influences the predictive power of each cluster. Thus, we examine two methods of selecting an optimal number of clusters.



The Within Cluster Sum of Squares (WCSS) measures the Euclidean distance from each point in the feature space to the centroid. We can plot this with respect to the number of clusters to identify an optimal "elbow" point to operate at.

We can see from the plot that the marginal improvement in WCSS decreases sharply around 5 clusters, so by the elbow method we would select $k = 5$ to be the optimal number of clusters.

7.5 Choosing Optimal Number of Clusters using Silhouette Method



Now, we attempt to find the optimal number of clusters k via the silhouette score, which is a score that measures the amount of overlap between clusters. A higher silouhette score implies that the cluster is easily separable from other clusters, while a lower silouhette score implies that a point in the cluster can potentially belong to multiple clusters. From the plot, we can see that $k = 3$, which corresponds to a silhouette score of 0.200, is the minimum.

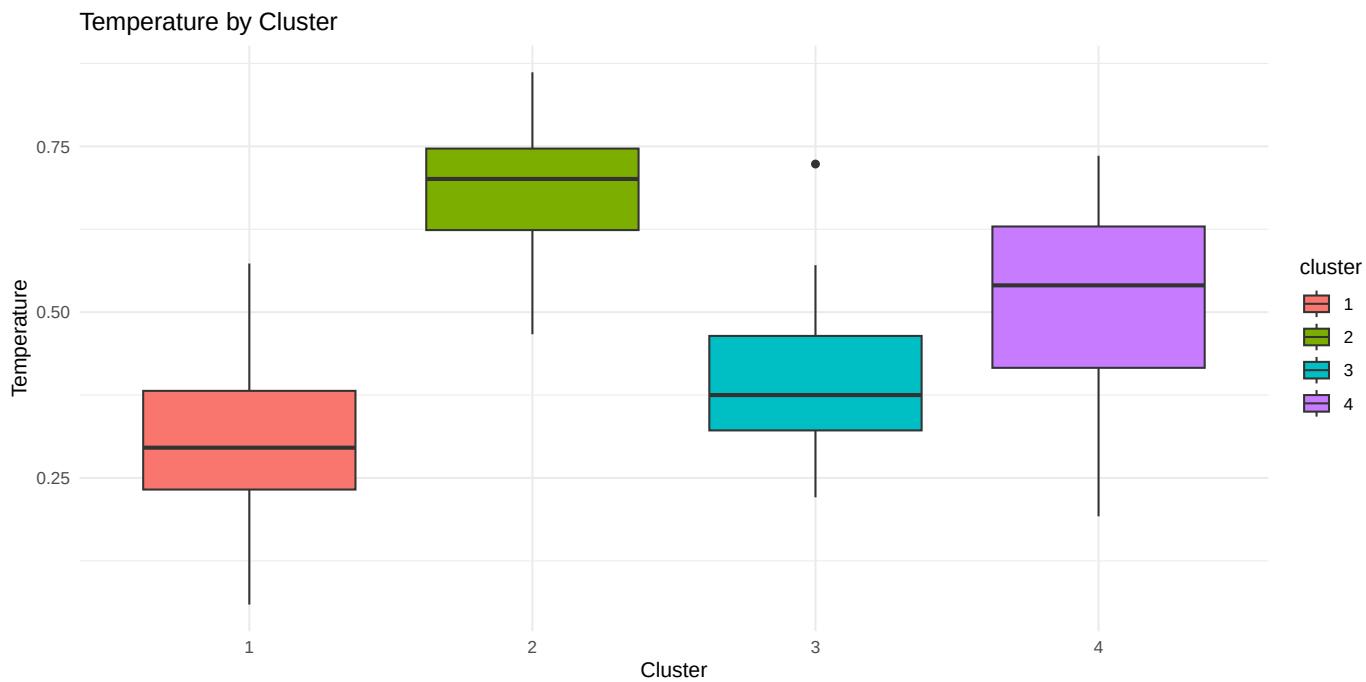
Interestingly, we can see that the k selected by the silhouette method is different than that selected for k means. Indeed, at $k = 3$, we can see that the marginal decrease to WCSS is still quite high. This may be because the silhouette and elbow methods are testing for criteria that are subtly different: silhouette is testing for the separability of the clusters, while elbow is testing for how compact the clusters are.

We thus pick an average of the two methods, $k = 4$ clusters, and refit the model on the following covariates: "temp", "atemp", "hum", "windspeed", "season", "weathersit", "weekday", and "workingday".

7.6 Distributions of Predictors within Each Cluster

Below, we have the distributions of each predictor with respect to each of the four clusters from the k-means algorithm trained on the previous section.

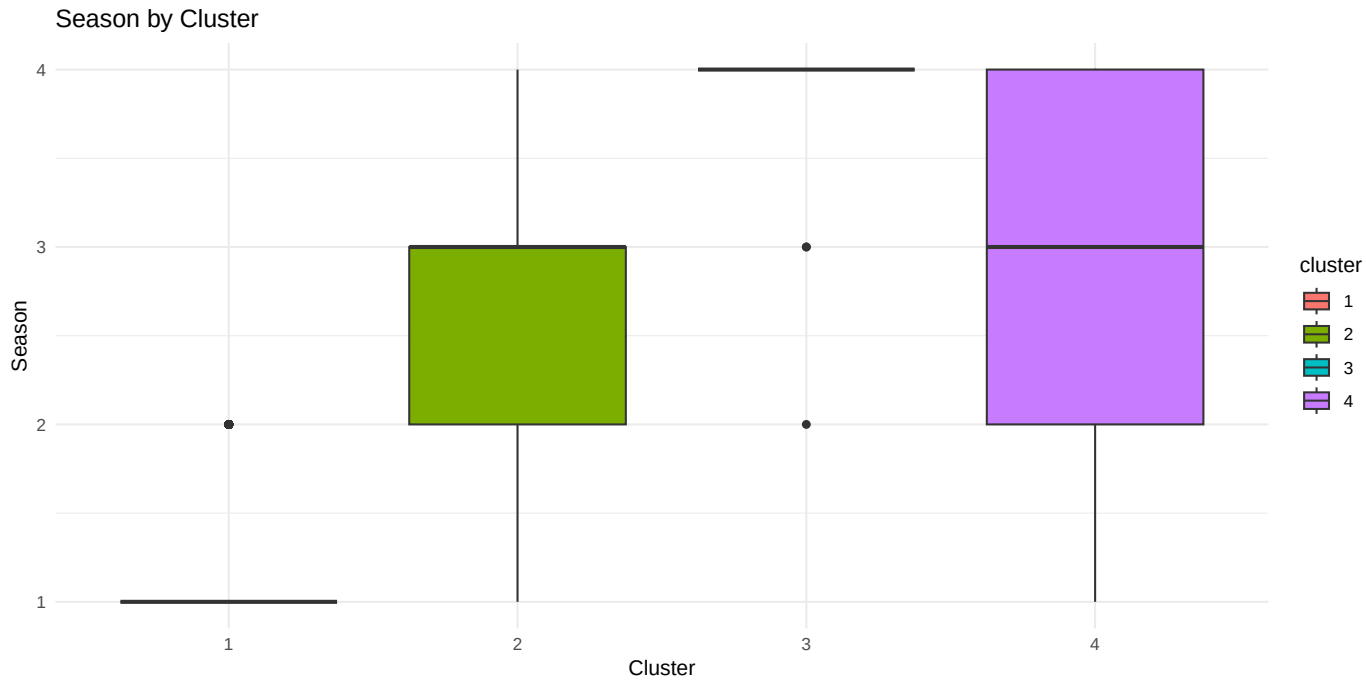
7.6.1 Distribution of Temperature by Cluster



The above boxplot displays the distribution of each cluster with respect to "temperature" for a k-means clustering with $k = 4$.

- We can see that the distributions (particularly the inner quartiles do not have significant overlap, suggesting that temperature was indeed a variable that was taken into consideration by the algorithm when constructing the four clusters.

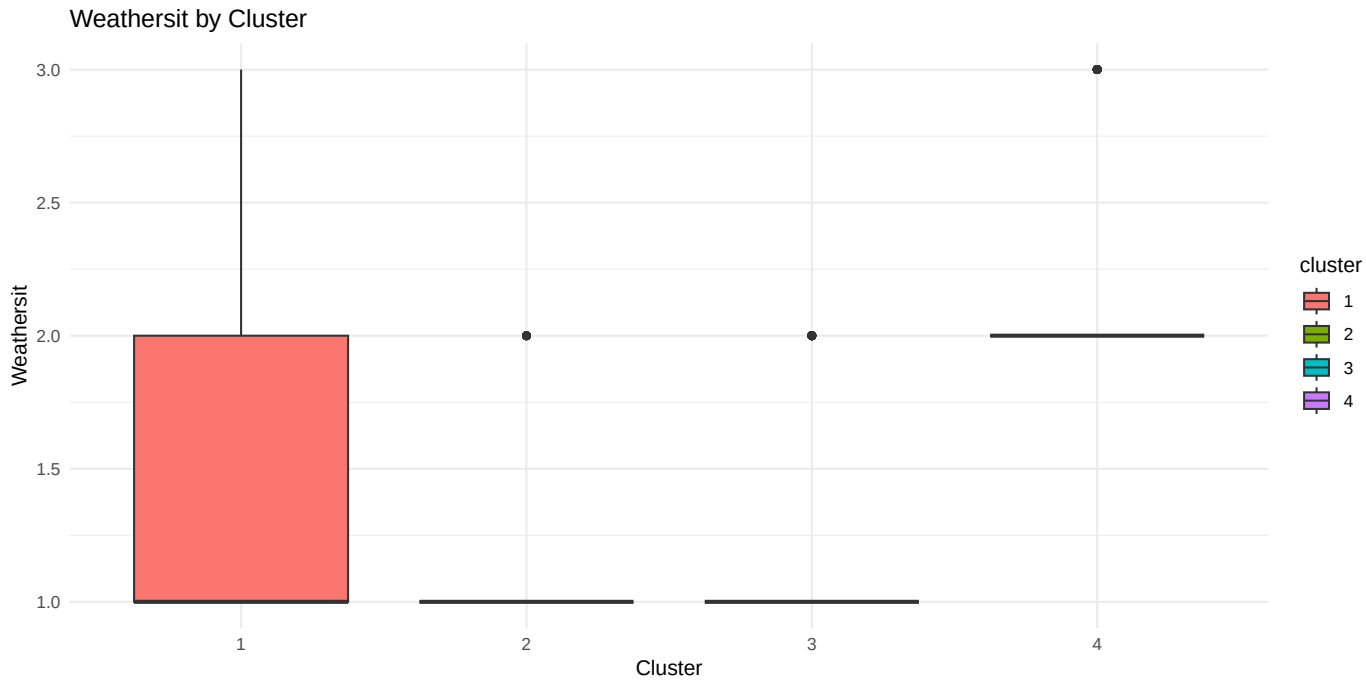
7.6.2 Distribution of Season by Cluster



The above boxplot displays the distribution of each cluster with respect to "season" for a k-means clustering with $k = 4$.

- We can see that cluster 1 exclusively selected bike rentals that took place in the spring, while cluster 3 exclusively selected bike rentals that took place in the winter (excluding outliers).
- However, the two clusters have a wider distribution. Cluster 4, for example, seemed to frequently have bike rentals that took place in spring, summer, and winter.

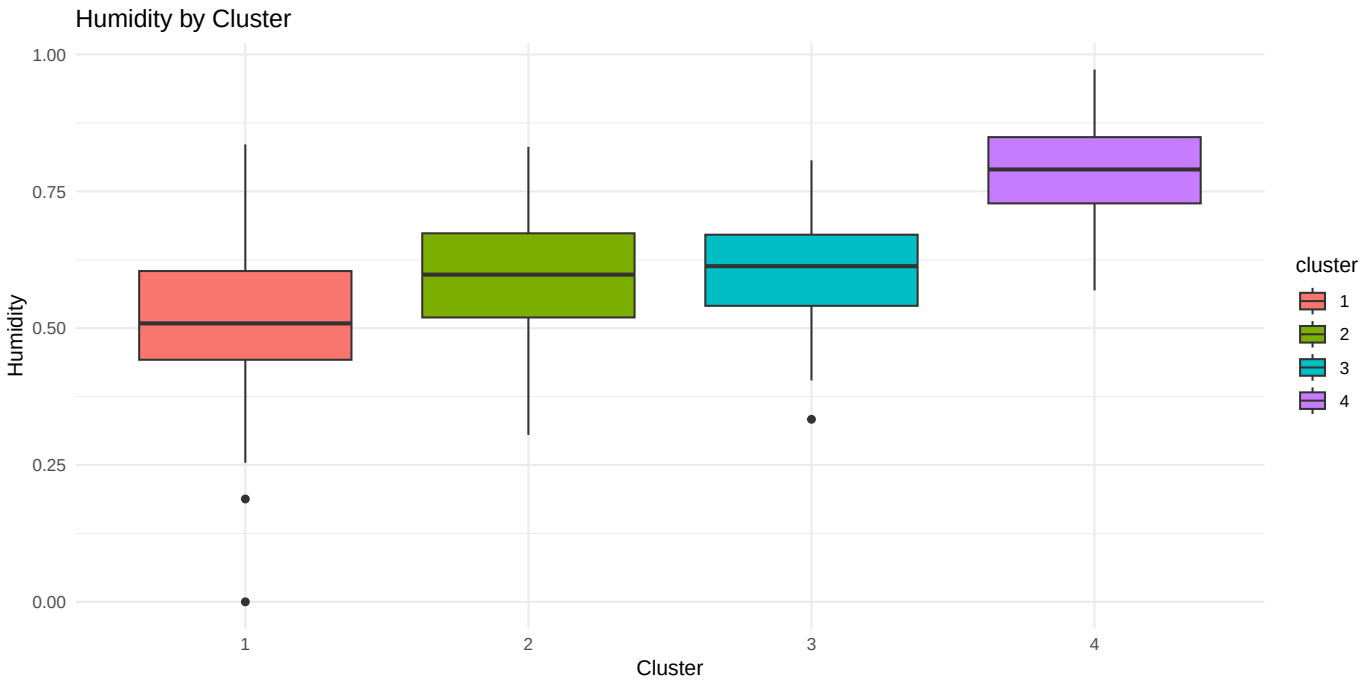
7.6.3 Distribution of Weathersit by Cluster



The above boxplot displays the distribution of each cluster with respect to "weathersit" for a k-means clustering with $k = 4$.

- Both clusters 2 and 3 almost exclusively selected observations from a "weathersit" value of 1.
- Cluster 4 seemed to exclusively select observations from a "weathersit" value of 4.
- The above seems to suggests that "weathersit" was an important predictor that greatly influenced the formation of the clusters.

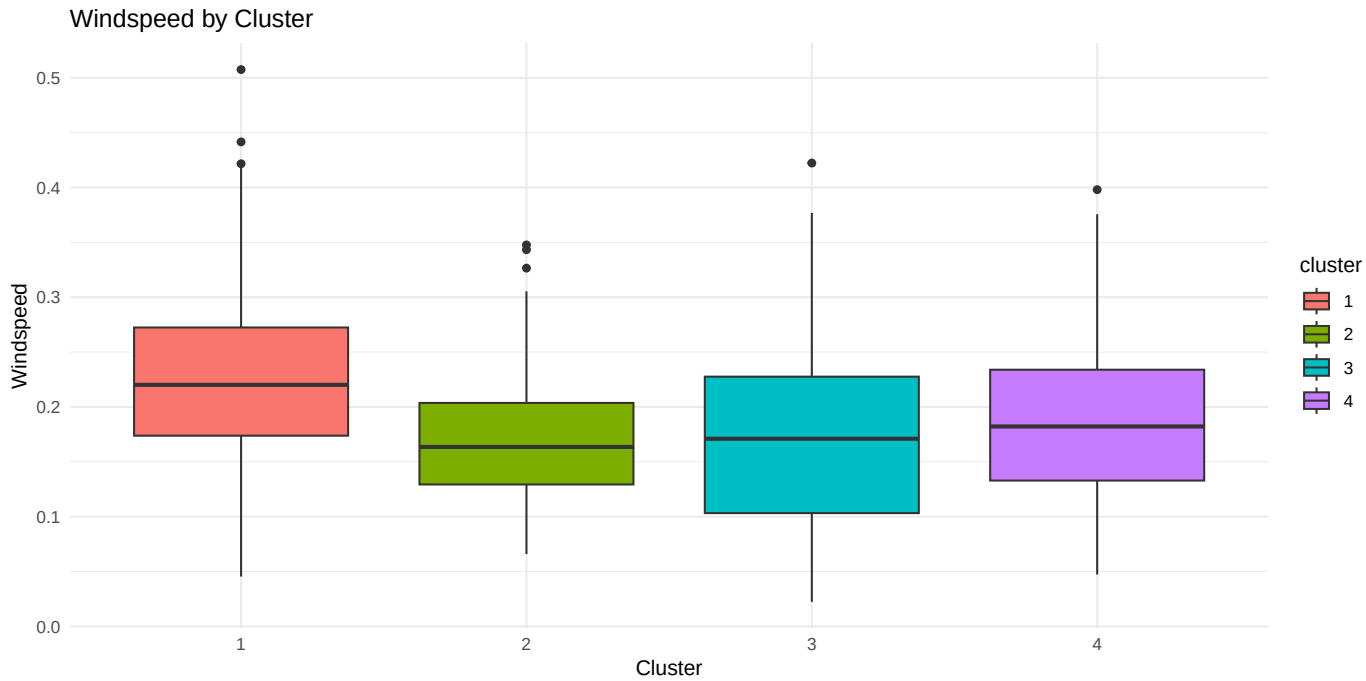
7.6.4 Distribution of Humidity by Cluster



The above boxplot displays the distribution of each cluster with respect to "humidity" for a k-means clustering with $k = 4$.

- We see significant overlap between the four clusters. Specifically, the inner quartiles of clusters 1, 2 and 3 seem to overlap with one another.
- Thus, we conclude that humidity may not have been a predictor that was greatly influential in the formation of the clusters.

7.6.5 Distribution of Windspeed by Cluster



The above boxplot displays the distribution of each cluster with respect to "windspeed" for a k-means clustering with $k = 4$.

- Like humidity, we can see that the inner quartiles of all four clusters overlap with one another. This suggests that windspeed may not have been an influential predictor for the formation of the k-means clusters.

8 Principal Component Analysis

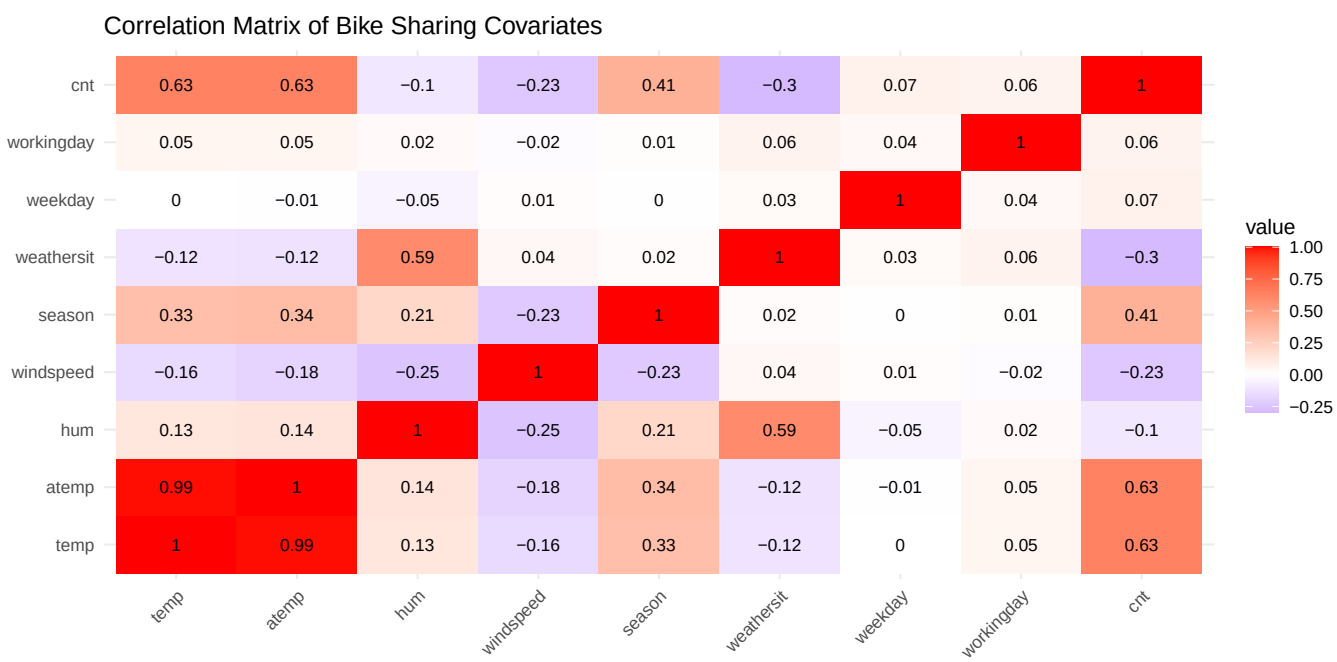
8.1 Motivation

We have seen through LASSO regression that we can reduce the number of covariates in the model and therefore reduce the feature space, a technique called dimension reduction. However, such a technique reduces the dimension by forcing the coefficient of that predictor to be 0 (i.e. there is no longer *any* contribution of that predictor to the model). Is it possible that we can do dimension reduction while still using important information from all relevant predictors?

Instead of simply sending coefficients to 0 as LASSO does, we can instead project each predictor in the original feature space to a smaller subspace. This technique is called Principal Component Analysis (PCA), and we hope to test its efficacy in predicting total bike rentals.

8.2 Evaluating Applicability of PCA Analysis

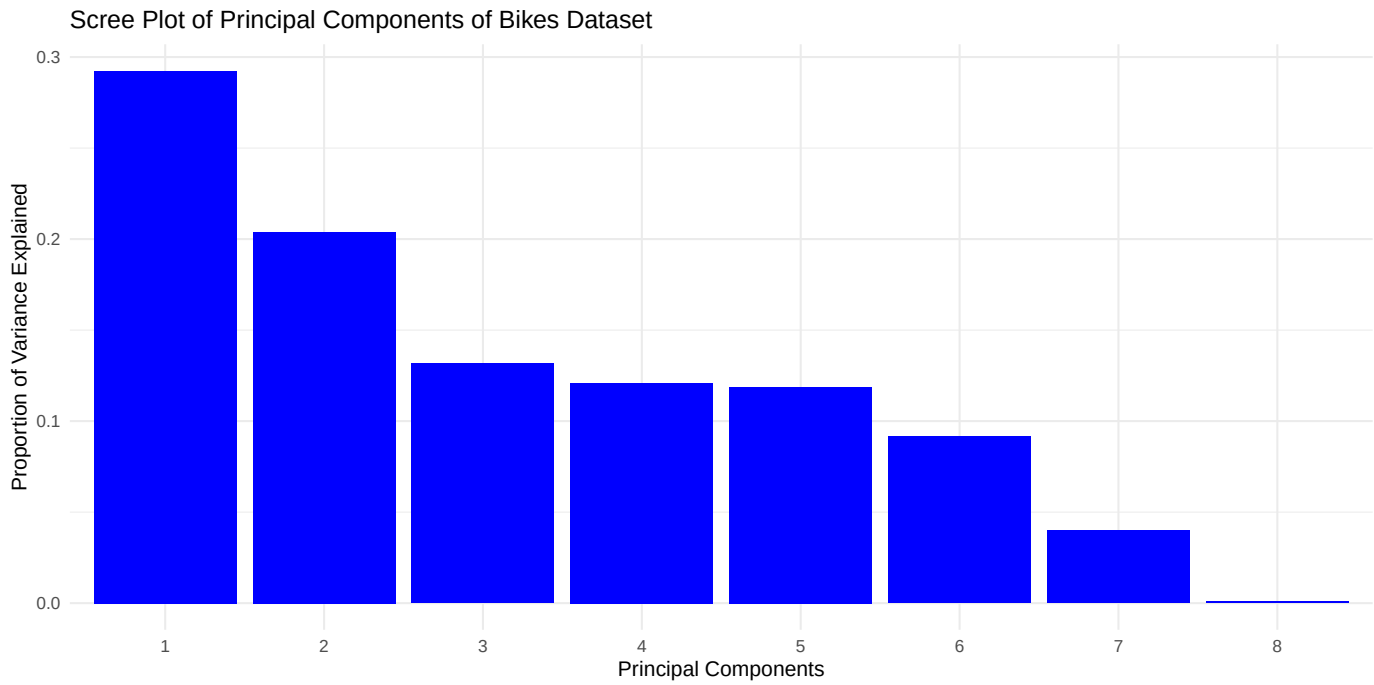
First, we must determine the applicability of PCA analysis on the `bikes` dataset. To do this, we plot a correlation matrix of all predictors. Higher correlations are a sign that dependence exists among predictors, and thus justifies the use of a dimensionality reduction technique such as PCA.



From the correlation matrix, we can see that most covariates in the dataset have little association with one another. However, there does seem to be some correlation among covariates related to weather and time, such as `season` and `temp` ($\rho = 0.33$), and `weathersit` and `hum` ($\rho = 0.59$).

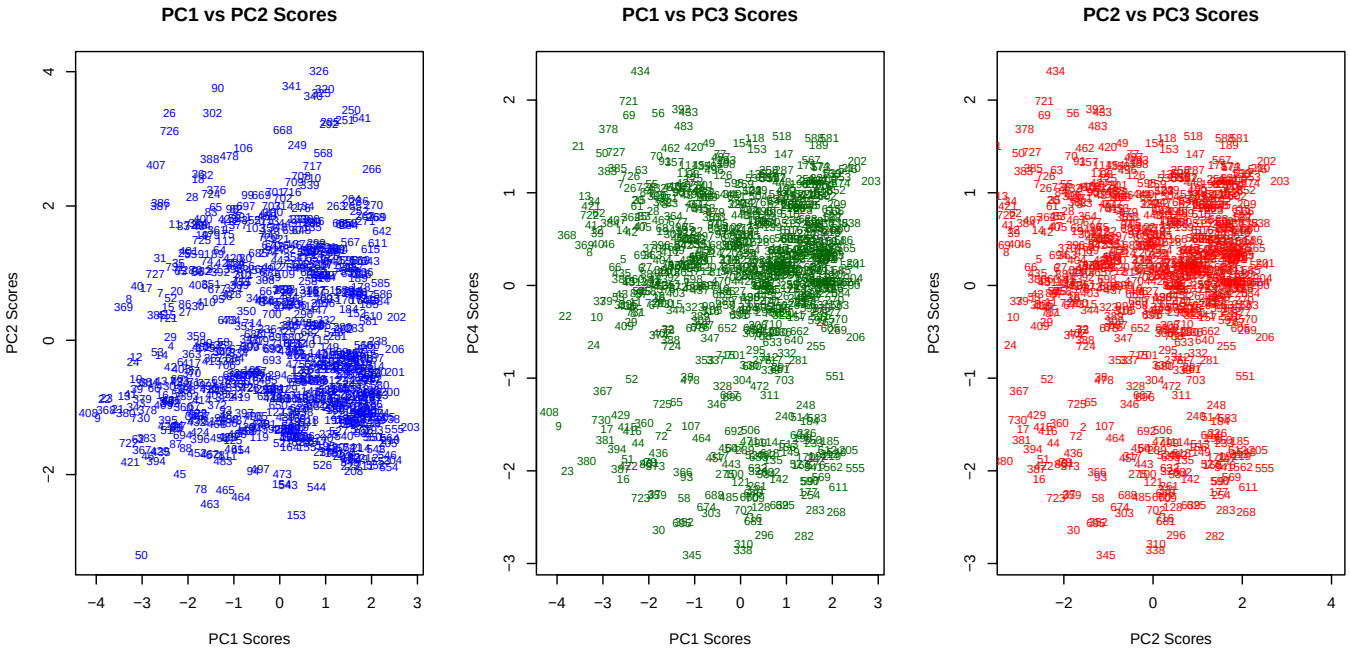
We can also see a moderate positive association between `season`, `temp` and the response variable `cnt`. Thus, as we have established several associations within predictors as well as predictors and response, we have sufficient motivation to run PCA analysis.

8.3 Choosing the Number of Principal Components



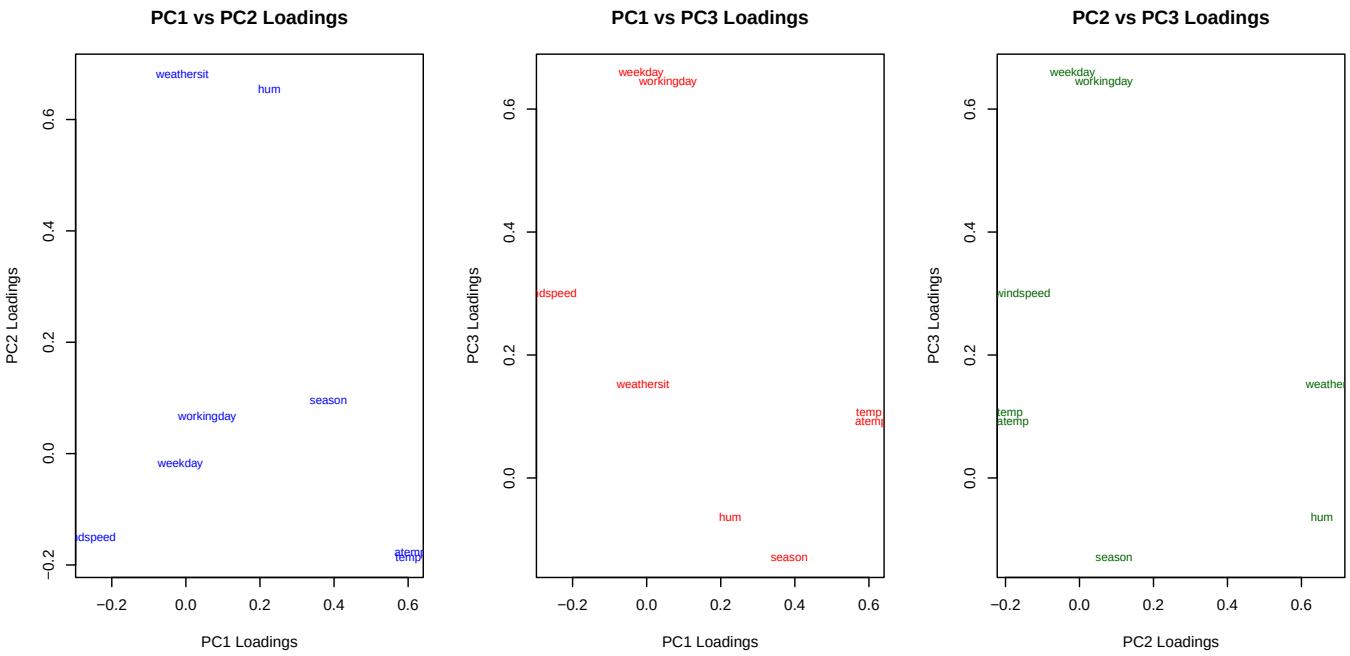
From the screeplot, we use the elbow method to determine that the optimal number of PCs (i.e. when the marginal decrease in variance explained decreases substantially) should be around 3.

8.4 Score Analysis of First Three Principal Components



The above PC score plots shows the projection of each day in the `bikes` dataset to a combination of two of the three principal components. We can see that the plots are overall fairly clustered, suggesting that most day are similar in covariate distribution. However, we can also see that several days have very high PC scores along a principal component; for example, there seems to be a significant number of days with PC2 scores greater than 2.

8.5 Loading Analysis of First Three Principal Components



Now we plot the principal component loadings of each covariate, which determines the influence of each covariate for a particular principal component.

- The first principal component seems to be centered around covariates that directly influence the temperature, with `temp`, `atemp`, and `season` having the greatest loadings for PC1.
- The second principal component seems to emphasize covariates that are related to the severity of the weather, such as `weathersit` and `humidity` having the greatest loadings along PC2.
- The third principal component seems to focus on covariates that determine the day of the week, such as `weekday` and `workingday`.

8.6 LASSO Regression onto Principal Components

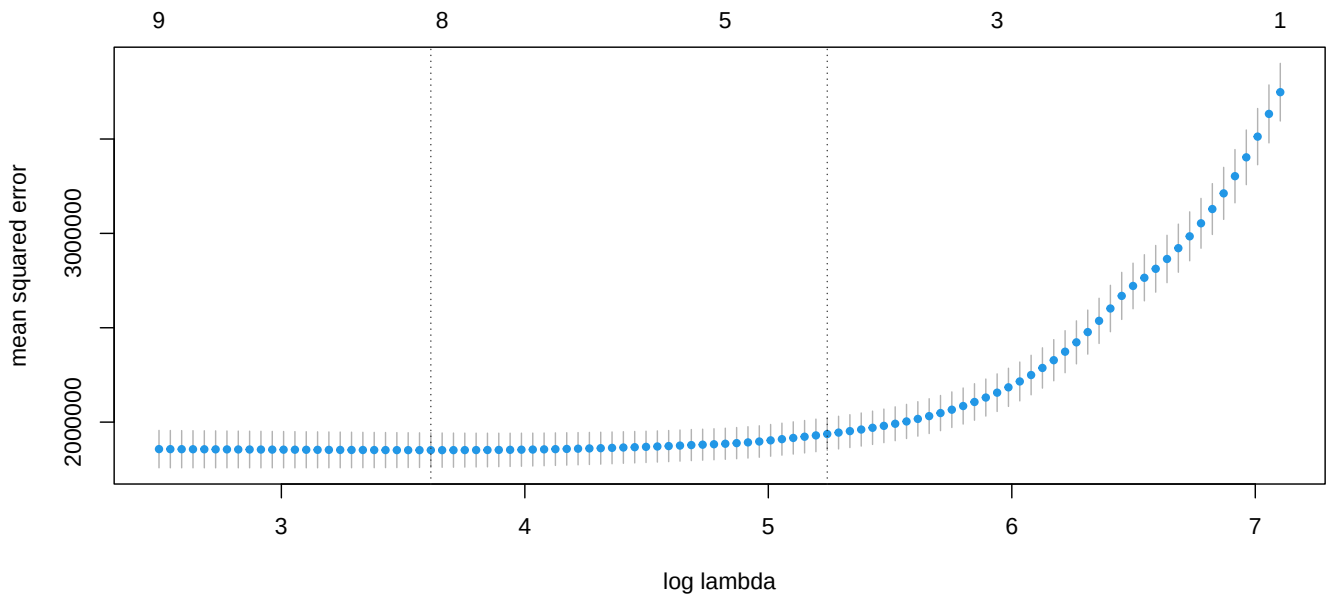


Table 15: LASSO Regression Coefficients for Principal Components

	Coefficient Value
intercept	4504.35
PC1	785.93
PC2	-491.99
PC3	66.36
PC4	169.37
PC5	93.61
PC6	140.88
PC7	167.40
PC8	0.00

```
Average magnitude of coefficients:  713.3205
```

From the table above, we can see that though we determined the optimal k to be 3 from the screeplot, LASSO regression determined that $k = 7$ principal components were significant, as seen by the non-zero coefficients.

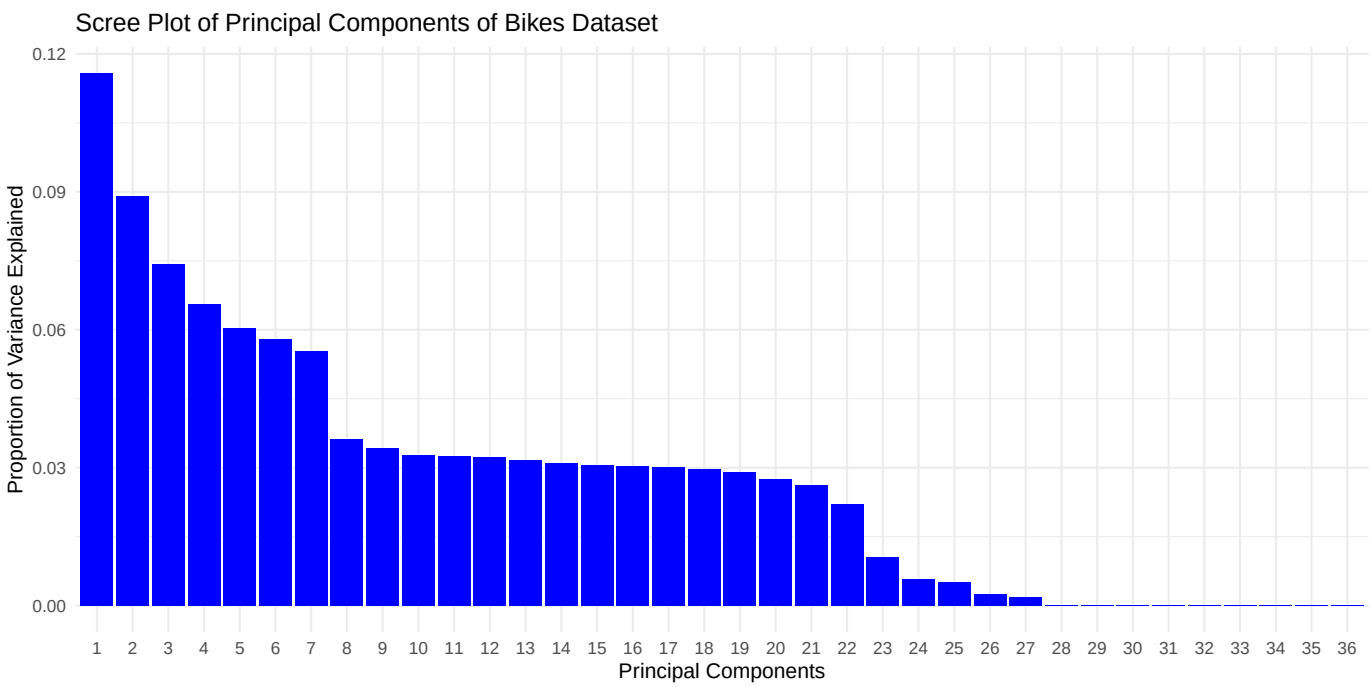
- This suggests that even though principal components 4 to 8 lead to a decrease in the marginal improvement in explained variance for the covariates, such components are still significant at predicting the number of bikes rented.

```
The OOS R^2 for CV LASSO Regression onto Principal Components is:  0.5066234
```

We then compute the R^2_{OOS} for LASSO regressed onto the principal components and get 0.5066. This is less than the R^2 for LASSO predicting total bike rental counts, regressed onto all covariates (0.8267).

- One reason for this is that in our PCA analysis, we only selected the covariates that had numeric values, and ignored the categorical predictors. Perhaps if we refit the PCA onto covariates on both the numeric values and categorical predictors converted into factors, we can reach a better OOS R^2.

8.7 PCA onto Categorical and Numerical Predictors



From the screeplot, we can see that the elbow point seems to be around $k = 8$, as there seems to be a sharp marginal decrease in the proportion of variance explained when we go from 8 to 9 principal components.

8.8 LASSO Regression onto Principal Components (Categorical + Numerical Predictors)

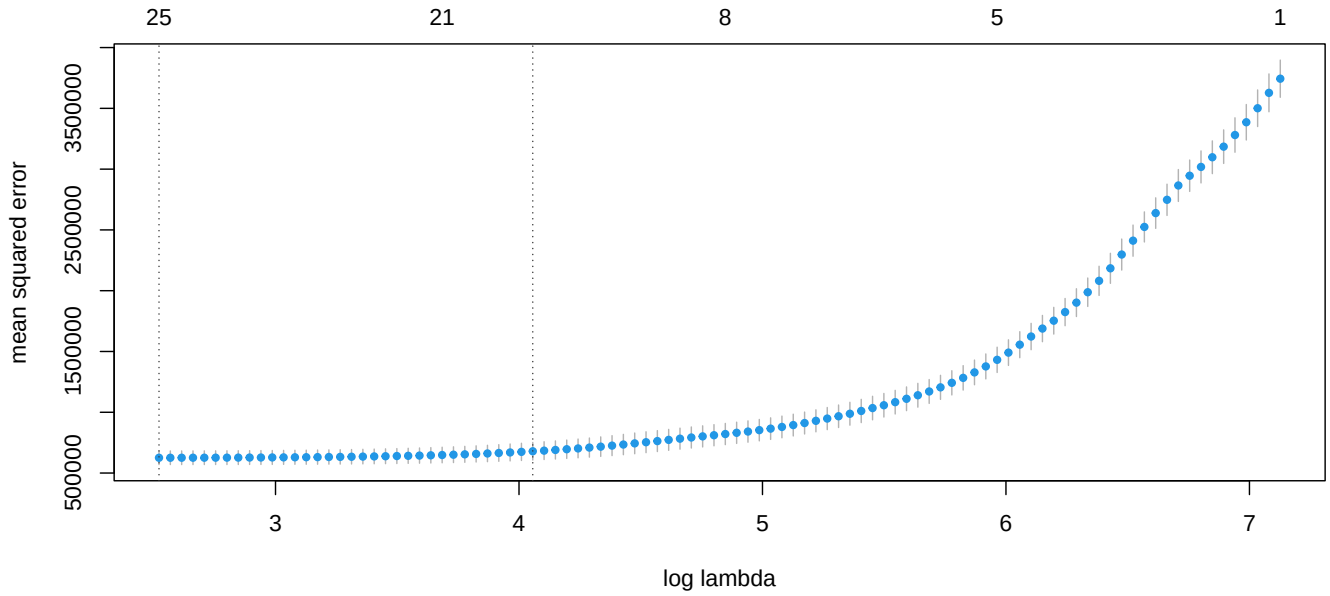


Table 16: LASSO Regression Coefficients for Principal Components

	Coefficient Value
intercept	4504.35
PC1	-602.07
PC2	64.71
PC3	56.75
PC4	-64.04
PC5	-296.22
PC6	476.72
PC7	575.03
PC8	-97.09
PC9	0.00
PC10	-64.94
PC11	-42.65
PC12	-97.54
PC13	8.88
PC14	26.32
PC15	86.86
PC16	-74.41
PC17	0.00
PC18	0.00
PC19	184.90
PC20	0.00

	Coefficient Value
PC21	250.92
PC22	-11.66
PC23	38.22
PC24	93.86
PC25	250.10
PC26	-177.58
PC27	0.00
PC28	0.00
PC29	0.00
PC30	0.00
PC31	0.00
PC32	0.00
PC33	0.00
PC34	0.00
PC35	0.00
PC36	0.00

Conducting CV LASSO regression on both categorical and numerical predictors results in a greater number of potential principal components (37 principal components for this model compared to 8 principal components when accounting for only numerical covariates).

- Out of the 37 principal components, we can see that CV LASSO has selected 23 principal components to be statistically significant ($\|\widehat{\beta}_i\| > 0, \forall i \in \{1, 37\}$).
- We can see that the average magnitude of coefficients for CV LASSO on both categorical and numerical predictors is smaller than that of CV LASSO on numerical predictors only (220.15 $<$ 713.32). This is likely due to the fact that the former model has more principal components than the latter, so the overall importance of each individual principal component is less.
- The OOS R^2 for the CV LASSO model regressed onto principal components constructed from categorical and numerical predictors is 0.8333, which is greater than the OOS R^2 for the CV LASSO model regressed onto principal components constructed from only numerical predictors (0.8333 $>$ 0.5066). Interestingly, the OOS R^2 of the current principal component model is even greater than the OOS R^2 of the CV LASSO model regressed onto the original covariates (0.833 $>$ 0.8267).

9 Regression Trees

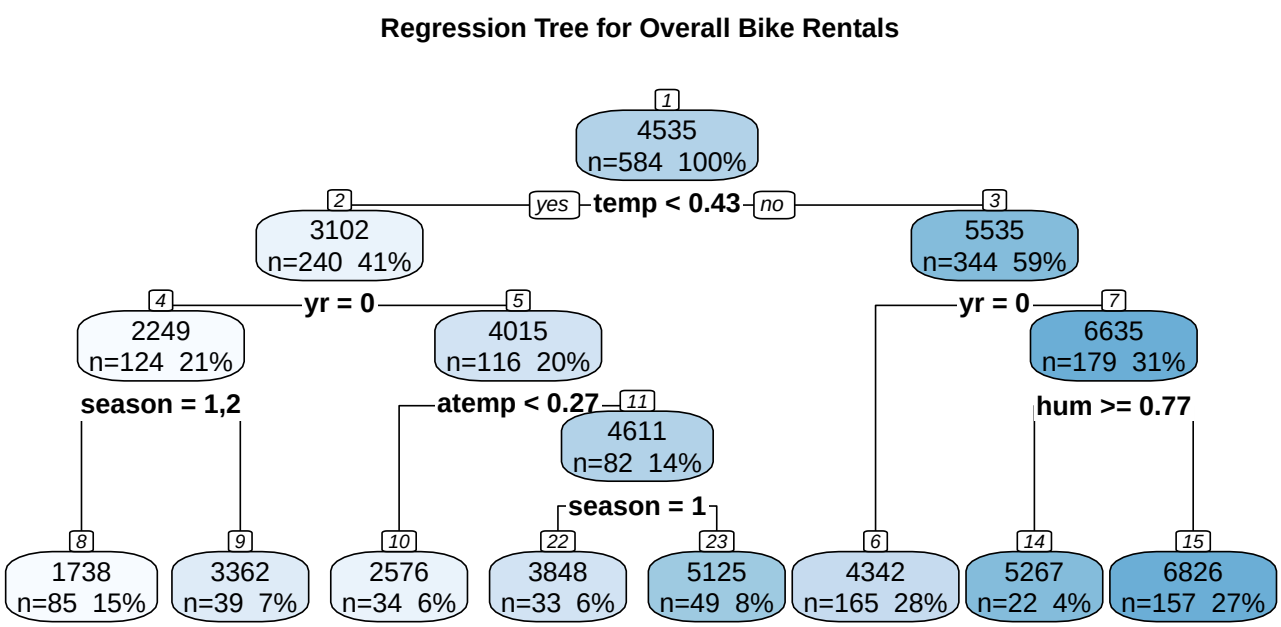
9.1 Motivation

So far, we have discussed naive and LASSO as our primary means of predicting the response variable from the original feature space. We have also introduced k-means clustering and principal component analysis as powerful tools that can help us fine-tune our prediction on the response by learning from trends in the predictors themselves. However, one central drawback of clustering and principal component analysis is that often times when we reduce the dimensions of the original feature space, we also lose the inherent meaning of the predictors themselves. For example, it can be difficult to understand what the first principal component or the first cluster intuitively means, as the first principal component is constructed from a linear combination of predictors in the original feature space. It can also be hard to intuitively understand what the first cluster represents, as various predictors may be considered important to that particular cluster.

Regression trees solve this issue by allowing us to split up the feature space into smaller subsets (as does k-means and PCA), but without directly aggregating different predictors together. Instead, regression trees partition the subspace into smaller segments based on predictors that are more similar to one another, and then lastly running a simpler regression on the remaining predictors. Thus, regression trees provide an intuitive yet powerful way of determining which predictors are most effective at predicting bike rentals.

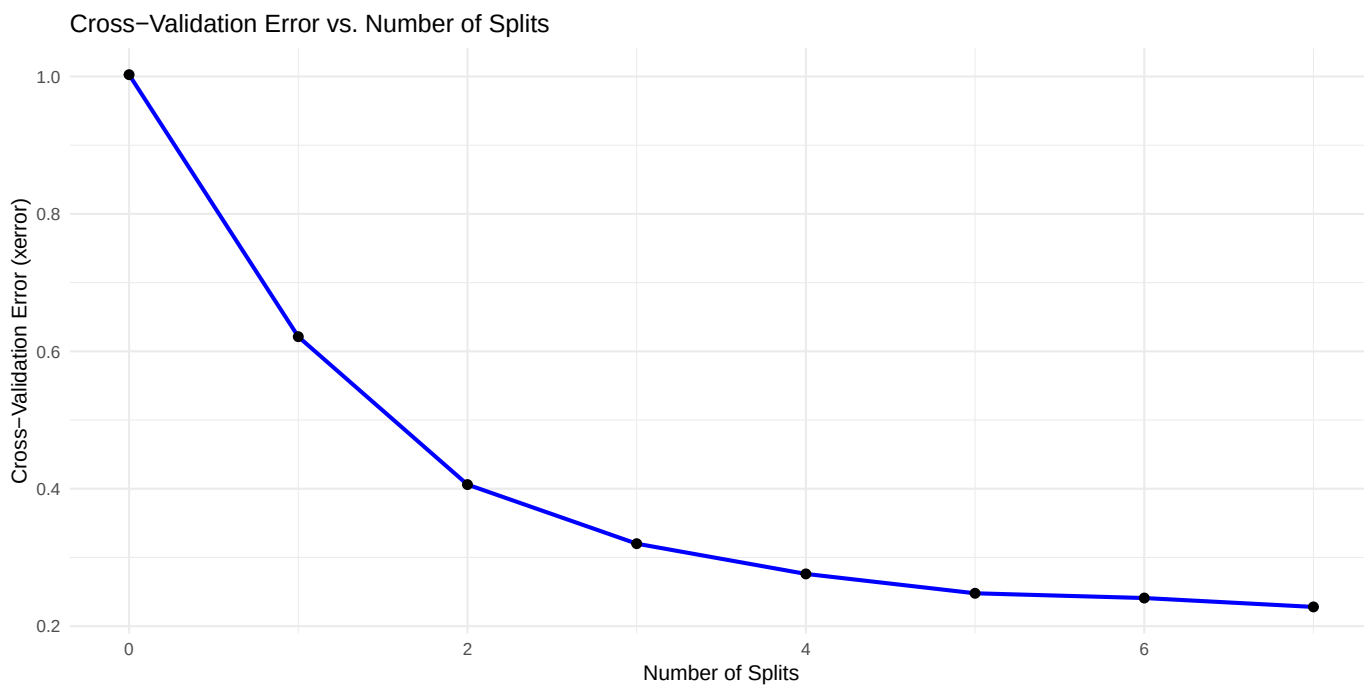
9.2 Regression Tree Analysis on Total Bike Rental Counts

We firstly run a regression tree on total bike rental counts. We use the following set of covariates for the prediction task: "season", "yr", "mnth", "weekday", "holiday", "workingday", "temp", "atemp", "hum", "windspeed", and "weathersit".



From the tree plot, we can make the following observations:

- From the original set of predictors, the regression tree model selected "temp", "yr", "season", and "hum" respectively to be the most important covariates for predicting the total number of bike rentals.
- We can see that the number of observations at each leaf (terminal) node is roughly evenly distributed. However, there are several nodes that have considerably more observations than others, in particular node 6 (observations with "temp"> 0.43, "yr" $= 0$) and node 15 (observations with "temp" > 0.43, "yr" $= 1$, "hum" < 0.77).



- From the line plot above, we can see that the number of splits that minimizes the CV error is seven splits (which corresponds to a final cross-validation error of 0.1944), which is consistent with the number of splits in our tree regression.

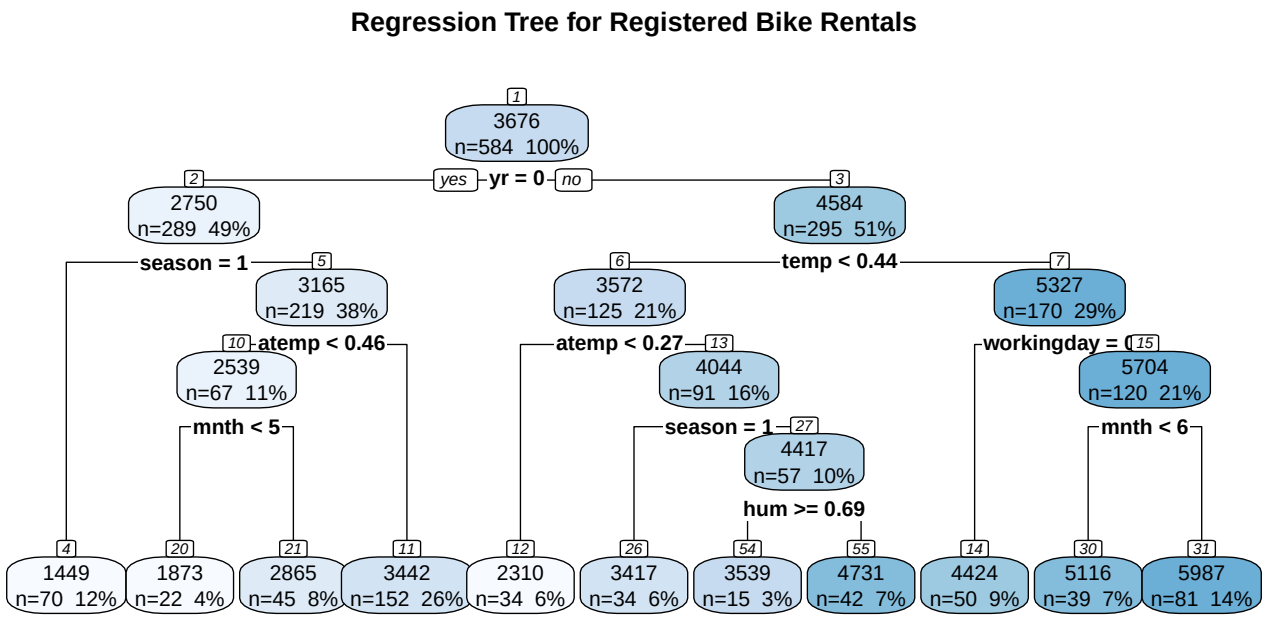
Table 17: Final MSE at Each Terminal Node

Node Index	Observations	Final MSE
8	22	2401877.4
9	49	777787.6
10	39	776470.8
22	157	723889.4
23	165	675084.4
6	34	643433.4
14	33	586557.2
15	85	335676.2

- From the table above, we can see that even though the number of observations in each leaf node are more or less similar (apart from the outliers discussed above), the MSEs vary quite greatly. This suggests that some decision rules in the tree partitioned the original feature space somewhat unevenly (as some leaf nodes have significantly lower MSE than others).

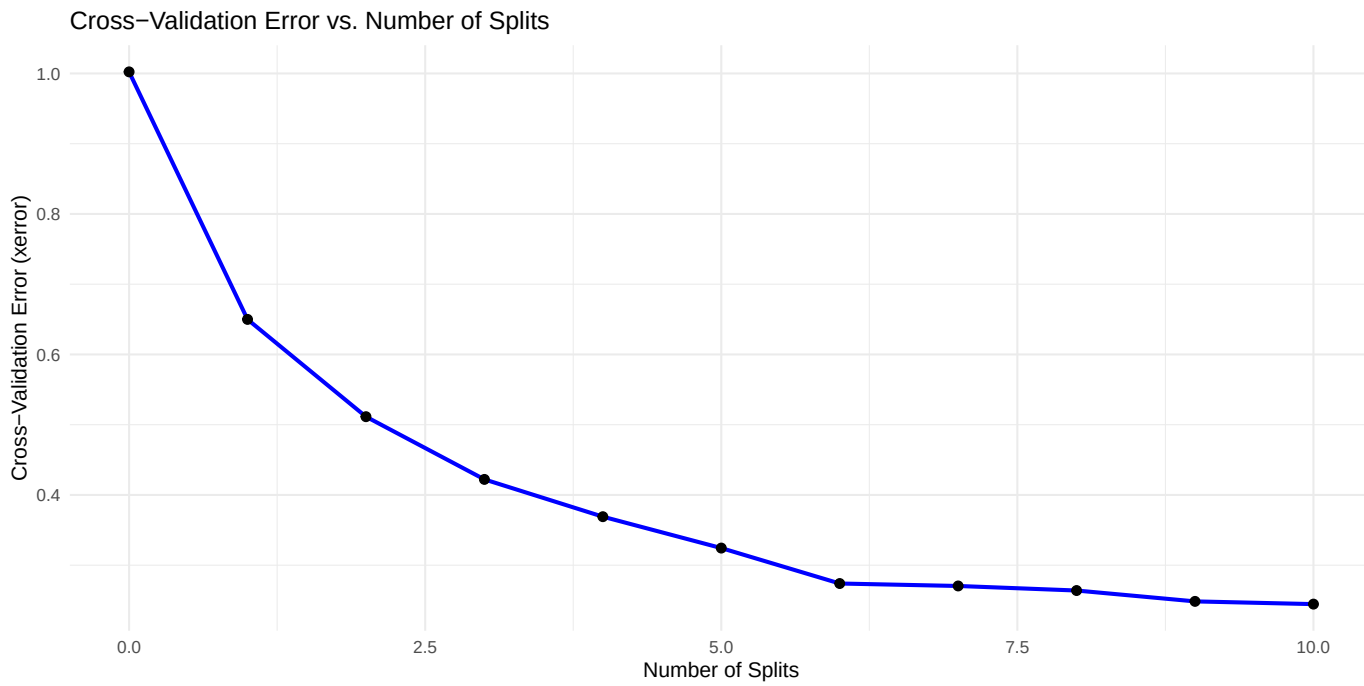
- The in-sample R^2 for the decision tree is 0.8056, which is fairly similar to the in-sample R^2 and OOS R^2 from naive and CV Lasso regression, respectively.

9.3 Regression Tree Analysis on Registered Bike Rental Counts



- From the tree plot, we can see that the regression model predicting the number of bike rental counts for registered users involved more splits than the regression model predicting total bike rental counts.

- The number of observations for the leaf nodes are overall lower than the leaf nodes from the previous tree regression; this is expected as the tree regression model for registered users involves more feature space partitions.



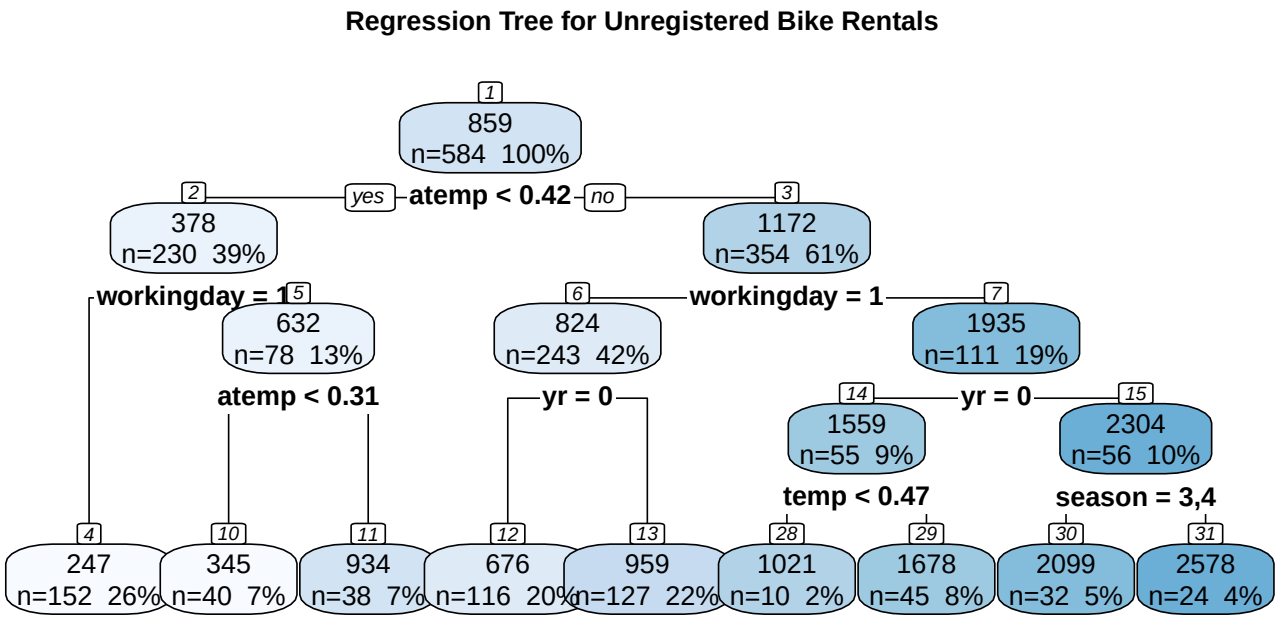
- From the line plot above, we can see that the number of splits that minimizes the CV error is 10 splits with a error of 0.2446. This is consistent with the number of splits in our tree regression model.

Table 18: Final MSE at Each Terminal Node

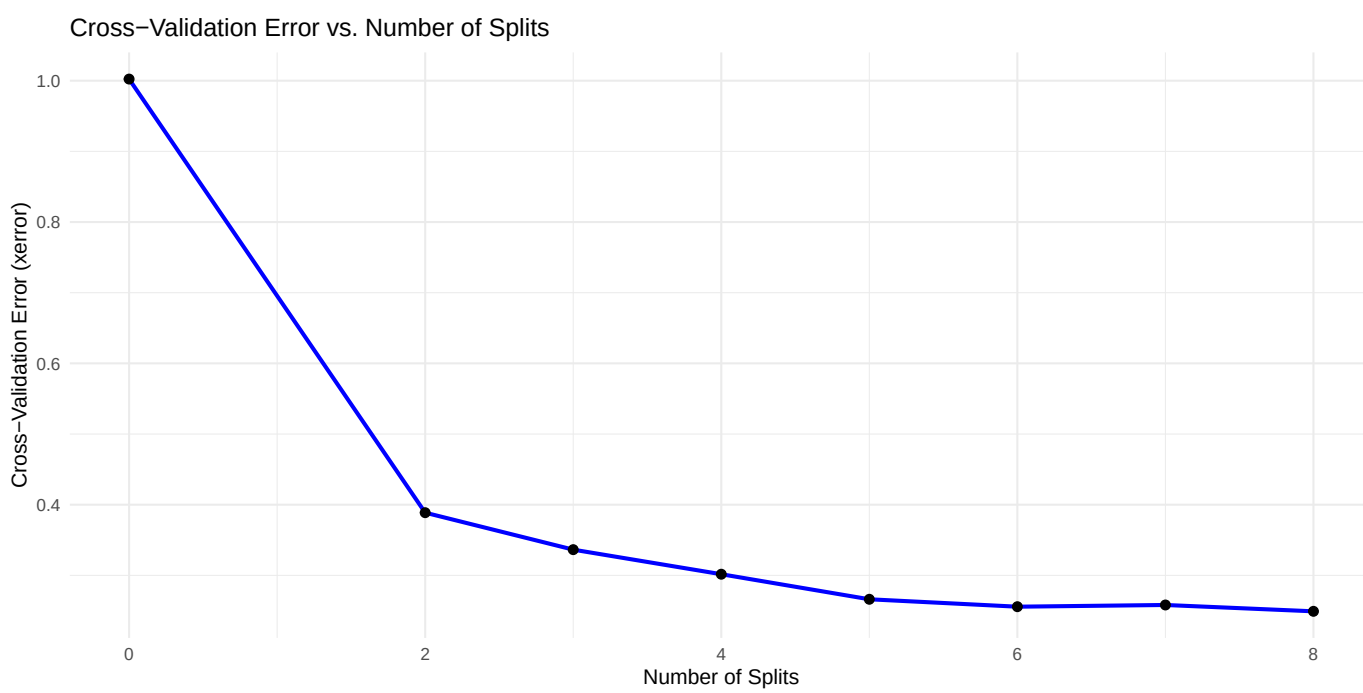
Node Index	Observations	Final MSE
4	15	1281956.9
20	42	699599.7
21	45	644785.5
11	34	606494.3
12	34	520578.1
26	39	493846.7
54	50	484584.6
55	81	433549.3
14	152	399187.5
30	22	298260.9
31	70	219198.8

- While there still does seem to be considerable amounts of variation in leaf node MSEs, the MSEs in general seem to smaller in this tree regression than compared to the tree regression predicting total bike rental counts. Specifically, we can see that nodes 30 and 31 have a MSE of below 300,000.
- Furthermore, we can also see that the distribution of MSE seems to be more uniform than the MSEs of the previous tree regression, as we now have 6 nodes with a MSE of below 500,000.
- The in-sample R^2 is 0.800, which is consistent to the R^2 values of previous models on the task of predicting bike rental counts from registered users.

9.4 Regression Tree Analysis on Unregistered Bike Rental Counts



- From the tree plot, we can see that the number of splits for the regression tree model of predicting the rental counts of unregistered users is roughly equal to the model of predicting the overall rental counts.
- Examining the distribution of observations at the leaf layer shows a more even distribution than the leaf layer observations for the previous two regression tree models. In particular, we can see that the leaf with the smallest number of observations ("atemp" < 0.42, "workingday" $= 1$) still is fairly large with 247 observations.



- From the line plot above, we can see that the number of splits that minimizes the CV error is 8 splits, which is consistent with the three regression model predicting the number of bike rentals for unregistered users.

Table 19: Final MSE at Each Terminal Node

Node Index	Observations	Final MSE
4	32	291825.84
10	24	252024.49
11	45	204198.76
12	10	180611.89
13	38	164924.81
28	127	80410.96
29	116	50955.62
30	40	49868.97
31	152	24236.23

- Interestingly, the MSEs for the tree regression predicting number of bike rentals for unregistered users are significantly lower than the MSEs of the previous two tree regressions (predicting total number of bike rentals and predicting number of bike rentals for registered users, respectively).
- Specifically, the root node only has a final MSE of 291825.84, which is the smallest MSE among the root nodes of the three models by a wide margin (the closest root node MSE comes from the second model at 1281956.9).
- The in-sample R^2 is 0.8041, which is a substantial improvement over the in-sample R^2 than the naive linear regression model on the task of prediction number of bike rentals for unregistered users.

10 Conclusion and Next Steps

Through this analysis, we have used a variety of techniques to determine how well each predictor in the dataset affects overall bike rental counts, bike rental counts among registered users, and bike rental counts among unregistered users. Specfically, we have analyzed the strength of each predictor primarily using regression techniques (naive, LASSO, and tree regression). We have also investigated possible dependencies between predictors through k-means clustering and PCA dimension reduction.

A possible next step in this analysis is to account for physical locations of the bike rental spots in overall bike rental counts. This can be done via the recording of a latitude and longitude coordinate of where a bike rental took place, and if the location of the bike rental is indeed significant, it can help Capital Rideshare determine optimal positions for future bike rental stations.

Another possible next step is to expand the study to multiple years. The current analysis only considers bike rentals from the years 2011-2012, so it may be difficult to capture chronological trends in the dataset. Extending the length of the study can perhaps determine long-term patterns in bike rentals that were not present in the current study.

11 Appendix

11.1 Selected Codes

Below is a portion of the code written for this report; for brevity, only the most important sections of code were included for reference.

11.1.1 Selected Code for Naive Regression

```
library(broom)

# Fit the full model
full_model <- lm(cnt ~ . - instant - dteday - casual - registered, data = bikes)

# Extract coefficients, standard errors, and p-values
model_results <- tidy(full_model)

# Select only necessary columns
model_table <- model_results[, c("term", "estimate", "std.error", "p.value")]
colnames(model_table) <- c("Coefficient", "Estimate", "Std. Error", "P-Value")

# Convert estimates & std errors to scientific notation
model_table$Estimate <- formatC(model_table$Estimate, format = "e", digits = 4)
model_table$`Std. Error` <- formatC(model_table$`Std. Error`, format = "e", digits = 4)
model_table$`P-Value` <- formatC(model_table$`P-Value`, format = "e", digits = 4)

# Display the table using kable()
kable(model_table, caption = "Standard Regression Coefficients with all Covariates among A

r_squared <- summary(full_model)$r.squared
```

```
# Fit the full model
full_model <- lm(registered ~ . - instant - dteday - cnt - casual, data = bikes)

# Extract coefficients, standard errors, and p-values
model_results <- tidy(full_model)

# Select only necessary columns
model_table <- model_results[, c("term", "estimate", "std.error", "p.value")]
colnames(model_table) <- c("Coefficient", "Estimate", "Std. Error", "P-Value")

# Convert estimates & std errors to scientific notation
model_table$Estimate <- formatC(model_table$Estimate, format = "e", digits = 4)
model_table$`Std. Error` <- formatC(model_table$`Std. Error`, format = "e", digits = 4)
model_table$`P-Value` <- formatC(model_table$`P-Value`, format = "e", digits = 4)
```

```
# Display the table using kable()
kable(model_table, caption = "Standard Regression Coefficients with all Covariates among P
```

11.1.2 Selected Code for LASSO Regression

```
library(gamlr)
# Load dataset
df <- read.csv("day.csv")

# Exclude 'dteday' and 'instant'
df <- df[, !(names(df) %in% c("casual", "registered", "dteday", "instant"))]

# Convert categorical variables to factors (important for LASSO)
df$season <- as.factor(df$season)
df$yr <- as.factor(df$yr)
df$mnth <- as.factor(df$mnth)
df$holiday <- as.factor(df$holiday)
df$weekday <- as.factor(df$weekday)
df$workingday <- as.factor(df$workingday)
df$weathersit <- as.factor(df$weathersit)

# Create feature matrix (X) and response vector (y)
X <- model.matrix(cnt ~ ., data = df)[, -1]  # Create design matrix, remove intercept
y <- df$cnt  # Response variable

# Perform cross-validated LASSO regression
cv_model <- cv.gamlr(X, y, family = "gaussian")

# Plot cross-validation error
plot(cv_model)

# Best lambda (penalty term)
best_lambda <- cv_model$lambda.min

# Coefficients at best lambda
lasso_coefs = coef(cv_model, select = "min")

# Convert to a data frame
lasso_coefs_df <- as.data.frame(as.matrix(lasso_coefs))

# Rename columns for clarity
colnames(lasso_coefs_df) <- c("Coefficient Value")

# Add PC names as a column
lasso_coefs_df$Predictor <- rownames(lasso_coefs_df)
```

```
# Reorder columns to display predictor names first
lasso_coefs_df <- lasso_coefs_df[, c("Predictor", "Coefficient Value")]

# Display the table using kable (without scientific notation)
kable(lasso_coefs_df, digits = 4, caption = "CV LASSO Regression Coefficients")
```

```
# Load dataset
df <- read.csv("day.csv")

# Exclude 'dteday' and 'instant'
df <- df[, !(names(df) %in% c("casual", "registered", "dteday", "instant"))]

# Convert categorical variables to factors (important for LASSO)
df$season <- as.factor(df$season)
df$yr <- as.factor(df$yr)
df$mnth <- as.factor(df$mnth)
df$holiday <- as.factor(df$holiday)
df$weekday <- as.factor(df$weekday)
df$workingday <- as.factor(df$workingday)
df$weathersit <- as.factor(df$weathersit)

d = bikes$season
X <- model.matrix(cnt ~ . - season, data = df)[, -1]  # Create design matrix, remove inter
cv.treat = cv.gamlr(X, d) #1st stage - regress d on x
dhat = predict(cv.treat$gamlr, X)
dhat = drop(dhat)

#Calculating in-sample R^2
SSE = sum((d - dhat)^2)
SST = sum((d - mean(d))^2)
in_sample_r2 <- 1 - SSE/SST

#Calulating OOS R^2
oos_mse <- min(cv.treat$cvm)
var_y <- var(d)
oos_r2 <- 1 - (oos_mse / var_y)

#Plotting d-hat vs d

ggplot(data = data.frame(d, dhat), aes(x = factor(d), y = dhat)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Boxplot of dhat by d",
       x = "d (Weather Condition)",
       y = "dhat (Predicted Values)") +
  theme_minimal()
```

```
#2-stage LASSO
causal <- gamlr(cbind(d,dhat,X), bikes$cnt, free=2) #2nd stage LASSO
coef_d <- coef(causal)["d",]

results_table <- data.frame(
  Metric = c("In Sample R²", "OOS R²", "Coefficient of d"),
  Value = c(in_sample_r2, oos_r2, coef_d)
)

# Print as a kable table
kable(results_table, col.names = c("Metric", "Value"), caption = "2-Stage LASSO Results")
```

11.1.3 Selected Code for Clustering

```
library(factoextra)

set.seed(123)  # Ensures reproducibility

df <- read.csv("day.csv")
df <- df[, c("atemp", "hum", "windspeed")]
df_scaled = scale(df)

# Apply K-Means
kmeans_model <- kmeans(df_scaled, centers = 3, nstart = 25)

fviz_cluster(kmeans_model, data = df_scaled, geom = "point") +
  labs(title = "K-Means on Projected Axes of Average Temperature, Humidity and Windspeed")

# Get cluster counts
cluster_counts <- as.data.frame(table(kmeans_model$cluster))

# Rename the columns
colnames(cluster_counts) <- c("Cluster", "Count")

# Format cluster names
cluster_counts$Cluster <- paste0("Cluster ", cluster_counts$Cluster)

# Display as a kable table
kable(cluster_counts, col.names = c("Cluster", "Number of Points"), caption = "Cluster Siz
```

```
library(cluster)

# Compute silhouette scores for different values of K
silhouette_scores <- sapply(2:10, function(k) {
  km <- kmeans(df_scaled, centers = k, nstart = 25)
```

```
  ss <- silhouette(km$cluster, dist(df_scaled))
  mean(ss[, 3])  # Extract average silhouette width
})

# Plot silhouette scores
plot(2:10, silhouette_scores, type = "b", pch = 19,
     xlab = "Number of Clusters (K)",
     ylab = "Average Silhouette Score",
     main = "Silhouette Method for Optimal K",
     ylim = c(0.2, 0.25))
```

11.1.4 Selected Code for Principal Component Analysis

```
numerical_features <- c("temp", "atemp", "hum", "windspeed",
                        "season", "weathersit", "weekday", "workingday")
df_numerical <- df[, numerical_features]

mypca = prcomp(df_numerical, scale=TRUE)

# Create Scree Plot Data
scree_data <- data.frame(PC = 1:length(mypca$sdev),
                         Variance = (mypca$sdev^2) / sum(mypca$sdev^2))

scree_data$PC <- as.factor(scree_data$PC)

# Plot Scree Plot with Custom Y-Axis Ticks
ggplot(scree_data, aes(x = PC, y = Variance)) +
  geom_bar(stat = "identity", fill = "blue") +
  labs(title = "Scree Plot of Principal Components of Bikes Dataset",
       x = "Principal Components",
       y = "Proportion of Variance Explained") +
  theme_minimal()
```

```
# Extract PC scores from PCA output
scores <- as.data.frame(mypca$x)

# Set up a 1-row, 3-column layout for plots
par(mfrow = c(1, 3))

# Plot PC1 vs PC2
plot(scores$PC1, scores$PC2, type = "n",
     xlab = "PC1 Scores", ylab = "PC2 Scores",
     main = "PC1 vs PC2 Scores")
text(scores$PC1, scores$PC2, labels = rownames(scores), col = "blue", cex = 0.7)
```

```
# Plot PC1 vs PC3
plot(scores$PC1, scores$PC3, type = "n",
     xlab = "PC1 Scores", ylab = "PC4 Scores",
     main = "PC1 vs PC3 Scores")
text(scores$PC1, scores$PC3, labels = rownames(scores), col = "darkgreen", cex = 0.7)

# Plot PC1 vs PC3
plot(scores$PC2, scores$PC3, type = "n",
     xlab = "PC2 Scores", ylab = "PC3 Scores",
     main = "PC2 vs PC3 Scores")
text(scores$PC1, scores$PC3, labels = rownames(scores), col = "red", cex = 0.7)
```

```
# Extract PC loadings (weights)
loadings <- as.data.frame(mypca$rotation)

# Set up a 1-row, 3-column layout for plots
par(mfrow = c(1, 3))

# Plot PC1 vs PC2
plot(loadings$PC1, loadings$PC2, type = "n",
     xlab = "PC1 Loadings", ylab = "PC2 Loadings",
     main = "PC1 vs PC2 Loadings")
text(loadings$PC1, loadings$PC2, labels = rownames(loadings), col = "blue", cex = 0.7)

# Plot PC1 vs PC3
plot(loadings$PC1, loadings$PC3, type = "n",
     xlab = "PC1 Loadings", ylab = "PC3 Loadings",
     main = "PC1 vs PC3 Loadings")
text(loadings$PC1, loadings$PC3, labels = rownames(loadings), col = "red", cex = 0.7)

# Plot PC2 vs PC3
plot(loadings$PC2, loadings$PC3, type = "n",
     xlab = "PC2 Loadings", ylab = "PC3 Loadings",
     main = "PC2 vs PC3 Loadings")
text(loadings$PC2, loadings$PC3, labels = rownames(loadings), col = "darkgreen", cex = 0.7
```

```
set.seed(123)
pc_scores <- as.data.frame(mypca$x)
y <- df$cnt
X <- as.matrix(pc_scores)
cv_lasso <- cv.gamlr(X, y, family = "gaussian")
plot(cv_lasso)

# Extract coefficients from the best LASSO model
lasso_coefs <- coef(cv_lasso$gamlr)  # Extract coefficients from cross-validated LASSO moc
```

```
# Convert to a data frame
lasso_coefs_df <- as.data.frame(as.matrix(lasso_coefs))

# Rename columns for clarity
colnames(lasso_coefs_df) <- c("Coefficient Value")

# Display table with kable()
kable(lasso_coefs_df, digits = 2, caption = "LASSO Regression Coefficients for Principal C

cat("Average magnitude of coefficients: ", mean(abs(lasso_coefs_df$`Coefficient Value`)))
```

11.1.5 Selected Code for Regression Trees

```
set.seed(123)
library(rpart)
library(rpart.plot)
# Load dataset
df <- read.csv("day.csv")

# Select relevant predictor variables
df <- df %>%
  select(cnt, season, yr, mnth, weekday, holiday, workingday, temp, atemp, hum, windspeed,

# Convert categorical variables to factors
df$season <- factor(df$season, levels = c(1, 2, 3, 4))
df$holiday <- factor(df$holiday, levels = c(0, 1))
df$workingday <- factor(df$workingday, levels = c(0, 1))
df$weathersit <- factor(df$weathersit, levels = c(1, 2, 3, 4))

# Split the data into training (80%) and testing (20%)
set.seed(123)  # Ensures reproducibility
train_indices <- sample(nrow(df), 0.8 * nrow(df))
train_data <- df[train_indices, ]
test_data <- df[-train_indices, ]

# Fit a regression tree model to predict bike rentals (cnt)
tree_model <- rpart(cnt ~ ., data = train_data, method = "anova")

rpart.plot(tree_model, type = 2, extra = 101, tweak = 1.2, main = "Regression Tree for Ove
```

```
# Load dataset
df <- read.csv("day.csv")

# Select relevant predictor variables
df <- df %>%
```

```
  select(registered, season, yr, mnth, weekday, holiday, workingday, temp, atemp, hum, wi

# Convert categorical variables to factors
df$season <- factor(df$season, levels = c(1, 2, 3, 4))
df$holiday <- factor(df$holiday, levels = c(0, 1))
df$workingday <- factor(df$workingday, levels = c(0, 1))
df$weathersit <- factor(df$weathersit, levels = c(1, 2, 3, 4))

# Split the data into training (80%) and testing (20%)
set.seed(123)  # Ensures reproducibility
train_indices <- sample(nrow(df), 0.8 * nrow(df))
train_data <- df[train_indices, ]
test_data <- df[-train_indices, ]

# Fit a regression tree model to predict bike rentals (cnt)
tree_model <- rpart(registered ~ ., data = train_data, method = "anova")

rpart.plot(tree_model, type = 2, extra = 101, tweak = 1.2, main = "Regression Tree for Re
```

```
# Load dataset
df <- read.csv("day.csv")

# Select relevant predictor variables
df <- df %>%
  select(casual, season, yr, mnth, weekday, holiday, workingday, temp, atemp, hum, windsp

# Convert categorical variables to factors
df$season <- factor(df$season, levels = c(1, 2, 3, 4))
df$holiday <- factor(df$holiday, levels = c(0, 1))
df$workingday <- factor(df$workingday, levels = c(0, 1))
df$weathersit <- factor(df$weathersit, levels = c(1, 2, 3, 4))

# Split the data into training (80%) and testing (20%)
set.seed(123)  # Ensures reproducibility
train_indices <- sample(nrow(df), 0.8 * nrow(df))
train_data <- df[train_indices, ]
test_data <- df[-train_indices, ]

# Fit a regression tree model to predict bike rentals (cnt)
tree_model <- rpart(casual ~ ., data = train_data, method = "anova")

rpart.plot(tree_model, type = 2, extra = 101, tweak = 1.2, main = "Regression Tree for Un
```